
Towards Understanding the Dynamics of Low-Rank Adaptation

Shu Ding^{1,2} Yang Peng^{1,2} Hangan Zhou^{1,2} Xinyu Lu^{1,2} Shangwei Chen^{1,2}
Junhua Huang³ Mingxuan Yuan³ Wei Wang^{1,2}

Abstract

Low-Rank Adaptation (LoRA) is a widely used parameter-efficient fine-tuning technique, and previous works have studied the update dynamics of LoRA, showing that updating via the low-rank matrix $\mathbf{A}$ confines the gradient $\nabla f(\mathbf{W})$ to the subspace spanned by $\mathbf{A}^\top \mathbf{A}$. However, few works have analyzed how the properties of the low-rank matrices affect the performance of LoRA, since existing methods heuristically initialize the low-rank matrices as Gaussian matrices. In this paper, we provide a theoretical understanding of the update dynamics of LoRA. We reveal that the update dynamics can be viewed as projecting the gradient $\nabla f(\mathbf{W})$ onto the subspace spanned by $\mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^\dagger \mathbf{A}$, and prove that when the gradient $\nabla f(\mathbf{W})$ is unavailable, if $\mathbf{A}$ is an Equiangular Tight Frame (ETF), $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^\dagger \mathbf{A}$ can preserve the maximum information from the gradient $\nabla f(\mathbf{W})$. Thus, initializing $\mathbf{A}$ as an ETF is the optimal solution for low-rank adaptation when the gradient $\nabla f(\mathbf{W})$ is unavailable. Furthermore, we establish the convergence of Low-Rank Adaptation with a rate of $\mathcal{O}(\frac{1}{T})$ when $\mathbf{A}$ is an ETF. Extensive experiments show that initializing the low-rank matrices as ETFs significantly outperforms the commonly used Gaussian initialization for existing primary LoRA variants.

1. Introduction

The rapid development of Large Language Models (LLMs) in recent years has established them as groundbreaking tools in real-world applications (OpenAI, 2023; Dubey et al., 2024; Comanici et al., 2025; DeepSeek-AI et al., 2025; Yang et al., 2025). However, the significant computational burden

of fine-tuning LLMs has necessitated the development of Parameter-Efficient Fine-Tuning (PEFT) techniques, among which Low-Rank Adaptation (LoRA) (Hu et al., 2022) has emerged as a widely used paradigm for PEFT. Motivated by the insight that over-parameterized models exhibit low intrinsic dimensions (Li et al., 2018; Aghajanyan et al., 2021), LoRA effectively adapts LLMs to downstream tasks by constructing two learnable low-rank matrices $\mathbf{A}$ and $\mathbf{B}$ to approximate model updates. Formally, LoRA decomposes the weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ into the formulation:

$$\mathbf{W} = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{B} \mathbf{A}, \quad (1)$$

where $\mathbf{W}_0 \in \mathbb{R}^{m \times n}$ is the pre-trained model weights, $\mathbf{B} \in \mathbb{R}^{m \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times n}$ are learnable low-rank matrices, $r \ll \min(m, n)$ is the rank, and α is a scaling factor.

While the insight of LoRA is simple, its update dynamics have not been fully understood. Prior works on LoRA have tried to explore the roles of LoRA matrices, revealing that fine-tuning $\mathbf{B}$ may be more effective than fine-tuning $\mathbf{A}$ (Hao et al., 2024; Zhu et al., 2024; Hayou et al., 2024). Consequently, subsequent works tend to focus on fixing $\mathbf{A}$ while updating $\mathbf{B}$. Hao et al. (2024) and He et al. (2025a) have stated that the training dynamics of LoRA can be viewed as confining the gradient $\nabla f(\mathbf{W})$ to the subspace spanned by $\mathbf{A}^\top \mathbf{A}$, and He et al. (2025a) further proposed a non-zero initialization for $\mathbf{B}$. However, $\mathbf{A}^\top \mathbf{A}$ is not a projection matrix for a general $\mathbf{A}$, thus the result of gradient compression highly depends on the singular values of $\mathbf{A}$. In practice, heuristic methods such as Gaussian initialization have been used to mitigate the influence of singular values.

Since LoRA performs gradient-based updates in the subspace $\mathbf{A}^\top \mathbf{A}$ induced by $\mathbf{A}$, the degree of information preserved by $\nabla f(\mathbf{W}) \mathbf{A}^\top \mathbf{A}$ highly depends on $\mathbf{A}$. If the gradient $\nabla f(\mathbf{W})$ is available, then performing SVD on $\nabla f(\mathbf{W})$ could derive the low-rank matrix $\mathbf{A}$ that preserves the maximum information (Zhang et al., 2025). However, LoRA aims for parameter-efficient updates by avoiding calculating the gradient $\nabla f(\mathbf{W})$, thus a critical yet challenging issue arises: *What properties should the low-rank matrices $\mathbf{A}$ and $\mathbf{B}$ have to preserve the maximum information of the unavailable gradient $\nabla f(\mathbf{W})$?* Few previous studies have touched this issue, while the commonly used Gaussian initialization

¹State Key Laboratory for Novel Software Technology, Nanjing University, China ²School of Artificial Intelligence, Nanjing University, China ³Huawei Noah's Ark Lab. Correspondence to: Wei Wang <wangw@lamda.nju.edu.cn>.

(Hu et al., 2022) cannot effectively preserve the information of the gradient $\nabla f(\mathbf{W})$ (see Section 3.2 for more details).

Furthermore, rigorous convergence analyses for LoRA remain limited. Malinovsky et al. (2024) provided theoretical analyses for LoRA with the assumption that the smallest eigenvalue of the induced projection matrix is positive, which implies that the rank of LoRA matrices equals the dimension of the pre-trained weight matrices. The similar issue also arises in (Guo et al., 2025) due to their reliance on the Frobenius inner product. Kim et al. (2025) provided theoretical analyses for LoRA with the assumption that the restricted strong convexity constant is positive. However, the constant becomes negative when the LoRA rank increases, limiting their scalability to higher ranks.

In this paper, we theoretically analyze the update dynamics of Low-Rank Adaptation. We derive the subspace induced by the LoRA update, prove that an Equiangular Tight Frame (ETF) structure for $\mathbf{A}$ can guarantee to preserve the maximum information of the gradient $\nabla f(\mathbf{W})$, and provide convergence analyses for LoRA, which are summarized as:

- We provide the Prototypical Low-Rank Adaptation in Algorithm 1 and show that the update dynamics of Algorithm 1 is $\mathbf{W}_{t+1} = \mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{P}_\mathbf{A}$. Here $\mathbf{P}_\mathbf{A} = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^\dagger \mathbf{A}$ is a projection matrix, which ensures that the projected gradient $\nabla f(\mathbf{W}_t) \mathbf{P}_\mathbf{A}$ is stable and avoids distortion caused by an ill-conditioned $\mathbf{A}$.
- We address the critical yet challenging issue of how to preserve the gradient information when the gradient is unavailable. We theoretically prove that when the gradient $\nabla f(\mathbf{W})$ is unavailable, the Equiangular Tight Frame (ETF) structure of $\mathbf{A}$ can guarantee to achieve the maximum lower bound for preserving the information of the gradient $\nabla f(\mathbf{W})$ (Theorem 3.3).
- We provide convergence analyses for Low-Rank Adaptation. Under the ETF initialization, we establish the convergence rate of $\mathcal{O}(\frac{1}{T})$ for the Prototypical Low-Rank Adaptation in Algorithm 1 for smooth loss functions (Theorem 3.4), and further establish the exponential convergence rate for smooth loss functions under the PL-condition (Theorem 3.6).
- Extensive experiments on real-world datasets show that initializing the low-rank matrices as ETFs significantly outperforms the commonly used Gaussian initialization for existing primary LoRA variants.

2. Related Works

LoRA, based on the assumption that over-parameterized models have low intrinsic dimensions (Li et al., 2018; Aghajanyan et al., 2021), is effective for adapting LLMs to downstream tasks by constructing two low-rank matrices $\mathbf{A}$ and

$\mathbf{B}$ to approximate model update (Hu et al., 2022). A few works have explored the structural modifications for LoRA to improve the fine-tuning performance (Liu et al., 2024; Kopiczko et al., 2024; Zhang et al., 2023; Tian et al., 2024; Yang et al., 2024; Yaras et al., 2024; Ding et al., 2023; Ren et al., 2024). Another line of works focus on subspace switching to enhance the learning capacity by stacking multiple LoRA modules in a chain-like manner (Lialin et al., 2024; Xia et al., 2024; Zhao et al., 2024; He et al., 2025b). However, these works lack theoretical analyses for the update dynamics of LoRA. Some studies on LoRA have explored the different roles of LoRA matrices, revealing that fine-tuning $\mathbf{B}$ may be more effective than fine-tuning $\mathbf{A}$ (Hao et al., 2024; Zhu et al., 2024). Hao et al. (2024) and He et al. (2025a) have stated that the training dynamics of LoRA can be viewed as confining the gradient $\nabla f(\mathbf{W})$ to the subspace spanned by $\mathbf{A}^\top \mathbf{A}$. Some subsequent works modify the learning rate (Hayou et al., 2024), rank (Valipour et al., 2023), and scaling factor (Kalajdzievski, 2023) to enhance the performance. However, $\mathbf{A}^\top \mathbf{A}$ is not a projection matrix for a general $\mathbf{A}$, thus the result of gradient compression highly depends on the singular values of $\mathbf{A}$.

Different from the commonly used Gaussian initialization (Hu et al., 2022), Hao et al. (2024) and He et al. (2025b) utilized the Stiefel manifold for initialization, Wang et al. (2024), Zhang et al. (2025) and Zhao et al. (2024) performed SVD on the gradient for initialization, Meng et al. (2024) and Wang et al. (2025a) performed SVD on the pre-trained model weights for initialization, He et al. (2025a) used the pseudo-inverse gradient compression for initialization. Nevertheless, performing SVD on the gradient $\nabla f(\mathbf{W})$ is impractical since LoRA aims for parameter-efficient updates by avoiding calculating the gradient. Moreover, Gaussian initialization and general Stiefel manifold elements cannot effectively preserve the unavailable gradient information. Intending to initialize $\mathbf{B}\mathbf{A}$ that approximates the gradient, Wang et al. (2025b), Yu et al. (2025) and Zhang & Pilanci (2024) heuristically derived the update rule with the form of imposing projection matrices on the gradient. However, these works do not analyze the properties of the LoRA matrices in preserving the information of the gradient $\nabla f(\mathbf{W})$.

Rigorous convergence analyses on LoRA remain limited. Malinovsky et al. (2024) provided theoretical analyses with the assumption that the smallest eigenvalue of the projection matrix is positive, which implies that the rank of the LoRA matrices equals the dimension of the pre-trained weights. A similar issue arises in (Guo et al., 2025) due to their reliance on the Frobenius inner product. Kim et al. (2025) provided convergence analyses with the assumption that the restricted strong convexity constant is positive. However, the constant becomes negative when the LoRA rank increases, making their analyses inapplicable at higher ranks.

3. Our Work

In the field of parameter-efficient fine-tuning (PEFT), the goal is to adapt a pre-trained large language model (LLM) to a downstream task. Given the model parameter $\mathbf{W}$, denote $f(\mathbf{W}) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ as the loss function. The objective of PEFT can be formulated as:

$$\min_{\Delta \mathbf{W} \in \mathbb{R}^{m \times n}} f(\mathbf{W}_0 + \Delta \mathbf{W}), \quad (2)$$

where $\mathbf{W}_0 \in \mathbb{R}^{m \times n}$ denotes the pre-trained model weights, and $\Delta \mathbf{W} \in \mathbb{R}^{m \times n}$ is the adaptation term to be updated. For LLMs, the total parameter dimension $m \times n$ is extremely large. Based on previous findings that pre-trained language models have low intrinsic dimensions (Li et al., 2018; Aghajanyan et al., 2021), Low-Rank Adaptation (LoRA) effectively adapts LLMs to downstream tasks by constructing two low-rank matrices $\mathbf{A}$ and $\mathbf{B}$ (Hu et al., 2022). Formally, the objective of LoRA is given by:

$$\mathbf{W} = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{B} \mathbf{A}, \quad (3)$$

where $\mathbf{B} \in \mathbb{R}^{m \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times n}$ are the learnable low-rank matrices, $r \ll \min(m, n)$ is the rank, and α is a scaling factor. Let $\nabla f(\mathbf{W}) \in \mathbb{R}^{m \times n}$ be the gradient of $f(\mathbf{W})$.

3.1. Update Dynamics of LoRA

In practice, LoRA updates the adaptation term by calculating the gradient with respect to $\mathbf{B}$ and $\mathbf{A}$. In the forward propagation process of LoRA, denote $\mathbf{W}_t = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$ as the parameter sequence, where $\mathbf{A}_t$ and $\mathbf{B}_t$ are the corresponding matrices at the t -th update round. Given the $\mathbf{A}$ matrix, the update rule for $\mathbf{B}$ with SGD is:

$$\mathbf{B}_{t+1} \leftarrow \mathbf{B}_t - \eta \nabla_{\mathbf{B}} f(\mathbf{W}_t), \quad (4)$$

where $\nabla_{\mathbf{B}} f(\mathbf{W}_t) = \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top$ and η is the learning rate. Similarly, given $\mathbf{B}$, the update rule for $\mathbf{A}$ with SGD is:

$$\mathbf{A}_{t+1} \leftarrow \mathbf{A}_t - \eta \nabla_{\mathbf{A}} f(\mathbf{W}_t), \quad (5)$$

where $\nabla_{\mathbf{A}} f(\mathbf{W}_t) = \mathbf{B}_t^\top \nabla f(\mathbf{W}_t)$. At the t -th round, the optimization resembles an alternating process, thus for a given $\mathbf{A}_t$, Hao et al. (2024) have demonstrated that the update dynamics of LoRA can be viewed as performing the following gradient compression:

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \frac{\alpha}{r} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top \mathbf{A}_t. \quad (6)$$

Similarly, for a given $\mathbf{B}_t$, the update dynamics of LoRA is:

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \frac{\alpha}{r} \mathbf{B}_t \mathbf{B}_t^\top \nabla f(\mathbf{W}_t). \quad (7)$$

However, for general low-rank matrices $\mathbf{A}_t$ or $\mathbf{B}_t$, $\mathbf{A}_t^\top \mathbf{A}_t$ in Eq. (6) or $\mathbf{B}_t \mathbf{B}_t^\top$ in Eq. (7) is not a projection matrix.

When it is used for gradient compression, the result depends strongly on the singular values of $\mathbf{A}_t$ or $\mathbf{B}_t$. The updates become too large or small in directions linked to the large or small singular values, which leads to unstable optimization when $\mathbf{A}_t$ or $\mathbf{B}_t$ is ill-conditioned. Moreover, guaranteeing the descent property of the loss, *i.e.*, $f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t)$, via gradient descent but without prior knowledge of the loss function imposes strict requirements on the learning rate.

The theoretical analyses on Low-Rank Adaptation based on gradient descent usually require prior knowledge of the loss function. The practical and commonly used function class is the L -smooth function, which is a standard setup in optimization theory (Boyd & Vandenberghe, 2004; Nesterov, 2004; Bottou et al., 2018; Nesterov et al., 2018; Khaled & Richtárik, 2023) and is widely adopted in LoRA-related research (He et al., 2025b; Malinovsky et al., 2024; Xia et al., 2024; Sun et al., 2024; Guo et al., 2025).

Assumption 3.1 (L -smoothness). The loss function f is differentiable and L -smooth, *i.e.*, there exists a constant $L \geq 0$ such that for any $\mathbf{W}, \mathbf{V} \in \mathbb{R}^{m \times n}$, the gradient is Lipschitz continuous with constant L :

$$\|\nabla f(\mathbf{W}) - \nabla f(\mathbf{V})\| \leq L \|\mathbf{W} - \mathbf{V}\|, \quad (8)$$

where $\|\cdot\|$ denotes the Frobenius matrix norm.

Consider a single update round in fine-tuning with $\mathbf{W}_{t+1} = \mathbf{W}_t + \Delta \mathbf{W}_t$, the following result holds for L -smooth loss functions (Boyd & Vandenberghe, 2004):

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) + \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t \rangle + \frac{L}{2} \|\Delta \mathbf{W}_t\|^2. \quad (9)$$

By minimizing the right hand side in Eq. (9), *i.e.*,

$$\min \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t \rangle + \frac{L}{2} \|\Delta \mathbf{W}_t\|^2, \quad (10)$$

the update term $\Delta \mathbf{W}_t$ that minimizes Eq. (10) takes the form $\Delta \mathbf{W}_t = -\frac{1}{L} \nabla f(\mathbf{W}_t)$. Consequently, the updated $\mathbf{W}_{t+1} = \mathbf{W}_t + \Delta \mathbf{W}_t$ ensures the descent property of the loss function, *i.e.*, $f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) - \frac{1}{2L} \|\nabla f(\mathbf{W}_t)\|^2$. The detailed derivation of the above result is deferred to Appendix A.1 due to space limitations.

A common technique to reduce the computational cost in fine-tuning is Low-Rank Adaptation (Hu et al., 2022), which approximates the update $\Delta \mathbf{W}_t$ via two low-rank matrices $\mathbf{A}_t$ and $\mathbf{B}_t$, where $\mathbf{B}_t \in \mathbb{R}^{m \times r}$ and $\mathbf{A}_t \in \mathbb{R}^{r \times n}$. Then the update rule takes the form $\mathbf{W}_{t+1} = \mathbf{W}_t + \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$, and $\Delta \mathbf{W}_t = \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$, therefore, Eq. (10) can be written as:

$$\min \left\langle \nabla f(\mathbf{W}_t), \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\rangle + \frac{L}{2} \left\| \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\|^2. \quad (11)$$

For a given $\mathbf{A}_t$, the gradient of Eq. (11) with respect to $\mathbf{B}_t$ takes the form:

$$\frac{\alpha}{r} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top + \frac{L\alpha^2}{r^2} \mathbf{B}_t \mathbf{A}_t \mathbf{A}_t^\top. \quad (12)$$

By setting the gradient in Eq. (12) equal to zero, the $\mathbf{B}_t$ that minimizes Eq. (11) takes the following form:

$$\mathbf{B}_t = -\frac{r}{L\alpha} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger, \quad (13)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. For an initialized $\mathbf{B}_t^0 = 0$, the term $\mathbf{B}_t$ can be derived via gradient descent, which is given by:

$$\begin{aligned} \mathbf{B}_t &= \mathbf{B}_t^0 - \frac{r}{L\alpha} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \\ &= \mathbf{B}_t^0 - \eta_B \nabla_{\mathbf{B}} f(\mathbf{W}_t) (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger, \end{aligned} \quad (14)$$

where $\nabla_{\mathbf{B}} f(\mathbf{W}_t) = \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top$ and $\eta_B = \frac{r}{L\alpha}$. In Eq. (14), $(\mathbf{A}_t \mathbf{A}_t^\top)^\dagger$ acts as a preconditioner that scales the gradient dimension-wise (Zhang & Pilanci, 2024; Yu et al., 2025). Thus, the update dynamics of LoRA takes the form:

$$\begin{aligned} \mathbf{W}_{t+1} &= \mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t \\ &= \mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}, \end{aligned} \quad (15)$$

where $\mathbf{P}_{\mathbf{A}_t} = \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t$. Symmetrically, for a given $\mathbf{B}_t$ and initialized $\mathbf{A}_t^0 = 0$, the term $\mathbf{A}_t$ takes the form:

$$\mathbf{A}_t = \mathbf{A}_t^0 - \eta_A (\mathbf{B}_t^\top \mathbf{B}_t)^\dagger \nabla_{\mathbf{A}} f(\mathbf{W}_t), \quad (16)$$

where $\nabla_{\mathbf{A}} f(\mathbf{W}_t) = \mathbf{B}_t^\top \nabla f(\mathbf{W}_t)$ and $\eta_A = \frac{r}{L\alpha}$, and the update dynamics of LoRA can be written as:

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \frac{1}{L} \mathbf{P}_{\mathbf{B}_t} \nabla f(\mathbf{W}_t), \quad (17)$$

where $\mathbf{P}_{\mathbf{B}_t} = \mathbf{B}_t (\mathbf{B}_t^\top \mathbf{B}_t)^\dagger \mathbf{B}_t^\top$. The detailed derivation is deferred to Appendix A.1 due to space limitations. We refer to the algorithm induced by the update dynamics in Eq. (15) and Eq. (17) as Prototypical Low-Rank Adaptation, and summarize the procedures in Algorithm 1. The subsequent analyses focus on the given $\mathbf{A}_t$ setting, and due to the symmetric structure of the two low-rank matrices $\mathbf{A}_t$ and $\mathbf{B}_t$, the similar results on $\mathbf{B}_t$ can be directly obtained.

Remark. Note that the update dynamics of Low-Rank Adaptation in Eq. (15) can be viewed as projecting the gradient $\nabla f(\mathbf{W})$ onto the subspace spanned by $\mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^\dagger \mathbf{A}$. In contrast to the term $\mathbf{A}_t^\top \mathbf{A}_t$ in Eq. (6), the matrix $\mathbf{P}_{\mathbf{A}_t} = \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t$ in Eq. (15) is a symmetric and idempotent projection matrix, i.e., the singular values of $\mathbf{P}_{\mathbf{A}_t}$ are restricted to 0 and 1, ensuring that the projected gradient $\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}$ is more stable during the updating process and avoids distortion caused by an ill-conditioned $\mathbf{A}_t$. It is worth noting that existing works have heuristically aligned the update rule in Eq. (6) with Eq. (15) by constraining the expected spectrum of $\mathbf{A}_t$ via Gaussian initialization (Hu et al., 2022) or Stiefel manifold (Hao et al., 2024; He et al., 2025b) to ensure $\mathbb{E}[\mathbf{A}_t \mathbf{A}_t^\top] = \mathbf{I}_r$. In these cases, $\mathbf{A}_t^\top \mathbf{A}_t$ approximates $\mathbf{P}_{\mathbf{A}_t}$ in expectation.

Algorithm 1 Prototypical Low-Rank Adaptation

Input: Pre-trained model $\mathbf{W}_0$, rank $r \ll \min\{m, n\}$, scaling factor α , update round T .

for $t = 0$ **to** $T - 1$ **do**

if Given the $\mathbf{A}_t$ matrix **then**

 Set the learning rate η_B and $\mathbf{B}_t^0 = 0$.

 Update $\mathbf{B}_t$: $\mathbf{B}_t \leftarrow \mathbf{B}_t^0 - \eta_B \nabla_{\mathbf{B}} f(\mathbf{W}_t) (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger$.

 Update $\mathbf{W}_{t+1} = \mathbf{W}_t + \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$.

else if Given the $\mathbf{B}_t$ matrix **then**

 Set the learning rate η_A and $\mathbf{A}_t^0 = 0$.

 Update $\mathbf{A}_t$: $\mathbf{A}_t \leftarrow \mathbf{A}_t^0 - \eta_A (\mathbf{B}_t^\top \mathbf{B}_t)^\dagger \nabla_{\mathbf{A}} f(\mathbf{W}_t)$.

 Update $\mathbf{W}_{t+1} = \mathbf{W}_t + \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$.

end if

end for

In the following, we discuss the relationship between the update rule in Eq. (15) and Eq. (17) and the practical implementation of LoRA, and address this from the perspective of the optimizer. When LoRA employs the AdamW (Loshchilov & Hutter, 2019) with a learning rate decay schedule as the optimizer (Hu et al., 2022), for a given $\mathbf{A}_t$, it optimizes $\mathbf{B}_t$ based on its gradient $\nabla_{\mathbf{B}} f(\mathbf{W}_t) = \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top$, while potentially applying a regularization term $\frac{\lambda}{2} \|\mathbf{B}_t\|^2$ during the optimization process (Loshchilov & Hutter, 2019). By setting the regularization operator as $\lambda = \frac{L\alpha^2}{r^2} \mathbf{A}_t \mathbf{A}_t^\top$, the regularization term becomes $\frac{\lambda}{2} \|\mathbf{B}_t\|^2 = \frac{L\alpha^2}{2r^2} \|\mathbf{B}_t\|_{\mathbf{H}}^2$ where $\mathbf{H} = \mathbf{A}_t \mathbf{A}_t^\top$ and $\|\cdot\|_{\mathbf{H}}$ is a weighted norm, which is equivalent to $\frac{L\alpha^2}{2r^2} \|\mathbf{B}_t \mathbf{A}_t^\top\|^2$. Optimizing through the gradient $\nabla_{\mathbf{B}} f(\mathbf{W}_t) = \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top$ with a specific learning rate $\eta = \frac{\alpha}{r}$ can be viewed as minimizing the term $\langle \nabla f(\mathbf{W}_t), \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \rangle$. Thus, optimizing with AdamW by setting $\lambda = \frac{L\alpha^2}{r^2} \mathbf{A}_t \mathbf{A}_t^\top$ can be viewed as minimizing the term $\langle \nabla f(\mathbf{W}_t), \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \rangle + \frac{L}{2} \|\frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t\|^2$, which is consistent with Eq. (11). This reveals that LoRA with the AdamW optimizer implicitly optimizes Eq. (11). The solution of Eq. (11) induces that the gradient descent operates within the projected subspace $\mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t$, thus the update dynamics of LoRA with AdamW can be viewed as in Eq. (15). In practice, existing works heuristically use Gaussian initialization (Hu et al., 2022) or Stiefel manifold (Hao et al., 2024; He et al., 2025b) to ensure $\mathbb{E}[\mathbf{A}_t \mathbf{A}_t^\top] = \mathbf{I}_r$, and the operator $\lambda = \frac{L\alpha^2}{r^2} \mathbf{A}_t \mathbf{A}_t^\top$ can be set as a constant.

3.2. ETF Guarantee Gradient Preservation

When LoRA is optimized via SGD, its update dynamics can be viewed as $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \frac{\alpha}{r} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top \mathbf{A}_t$; when LoRA is optimized via AdamW, its update dynamics can be viewed as $\mathbf{W}_{t+1} = \mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t$. In both cases, LoRA is optimized within the subspaces induced by $\mathbf{A}_t$, therefore, the degree of information preservation of the gradient $\nabla f(\mathbf{W}_t)$ is determined by $\mathbf{A}_t$. When the ma-

trix $\mathbf{A}_t$ is full-rank, $\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}$ retains all information from the gradient $\nabla f(\mathbf{W}_t)$ since the singular values of $\mathbf{P}_{\mathbf{A}_t}$ are 1, while $\nabla f(\mathbf{W}_t) \mathbf{A}_t^\top \mathbf{A}_t$ scales the information from $\nabla f(\mathbf{W}_t)$ by the singular values of $\mathbf{A}_t$. However, in Low-Rank Adaptation, the low-rank structure of $\mathbf{B}_t \mathbf{A}_t$ cannot capture all the information of the gradient $\nabla f(\mathbf{W}_t)$ with $r \ll \min\{m, n\}$. If the gradient $\nabla f(\mathbf{W}_t)$ is known, then constructing the projection matrix $\mathbf{P}_{\mathbf{A}_t}$ by performing Singular Value Decomposition (SVD) on $\nabla f(\mathbf{W}_t)$ could preserve its maximum information for low-rank matrix $\mathbf{A}_t$ (Zhang et al., 2025). Nevertheless, the key idea of LoRA is to reduce computational cost by avoiding calculating the gradient $\nabla f(\mathbf{W}_t)$. Existing works heuristically use Gaussian initialization (Hu et al., 2022) or Stiefel manifold (Hao et al., 2024; He et al., 2025b) with $\mathbb{E}[\mathbf{A}_t \mathbf{A}_t^\top] = \mathbf{I}_r$. Unfortunately, these initializations cannot effectively preserve the information of $\nabla f(\mathbf{W}_t)$. Here we provide an example for interpretation. Consider $\mathbf{A}_t$ consisting of r rows of an $n \times n$ orthogonal matrix, thus $\mathbf{A}_t$ belongs to the Stiefel manifold. $\mathbf{A}_t^\top \mathbf{A}_t = \mathbf{U}^\top (\mathbf{I}_r \oplus \mathbf{0}_{n-r}) \mathbf{U}$, where $\mathbf{U}$ is an orthogonal matrix and $\oplus$ denotes the direct sum. If the non-zero components of $\nabla f(\mathbf{W}_t)$ concentrate in the last $n-r$ coordinates, then $\|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\| = \|\nabla f(\mathbf{W}_t) \mathbf{A}_t^\top \mathbf{A}_t\| = 0$, which indicates complete energy loss.

How to preserve the maximum information of the gradient $\nabla f(\mathbf{W}_t)$ using low-rank matrices when the gradient $\nabla f(\mathbf{W}_t)$ is unavailable? This is a challenging yet critical issue in current research on LoRA. We address this issue by providing Theorem 3.3, which reveals that when the gradient $\nabla f(\mathbf{W}_t)$ is unavailable, the Equiangular Tight Frame (ETF) structure can achieve the maximum lower bound for gradient information preservation. Before proceeding to the theorem, we introduce the concept of ETF as follows:

Definition 3.2 (Equiangular Tight Frame (ETF)). A matrix $\mathbf{X} \in \mathbb{R}^{r \times n}$ is called an *equiangular tight frame (ETF)* if $\mathbf{X}$ satisfies: (1) each column has unit norm: $\|\mathbf{X}_{:,i}\|_2 = 1, \forall i \in \{1, \dots, n\}$; (2) the inner products between distinct columns have constant magnitude: $|\mathbf{X}_{:,i}^\top \mathbf{X}_{:,j}| = \sqrt{\frac{n-r}{r(n-1)}}, \forall i \neq j$; (3) $\mathbf{X} \mathbf{X}^\top = \frac{r}{n} \mathbf{I}_r$. Specifically, if $r = n$, then $\mathbf{X} = \mathbf{I}$. A matrix $\bar{\mathbf{X}} \in \mathbb{R}^{r \times n}$ is called a *normalized ETF* if $\bar{\mathbf{X}} = \sqrt{\frac{r}{n}} \mathbf{X}$ where $\mathbf{X}$ is an ETF, i.e., $\bar{\mathbf{X}} \bar{\mathbf{X}}^\top = \mathbf{I}_r$.

ETFs are a class of matrices with optimal properties in terms of coherence and tightness (Strohmer & Heath, 2003; Sustik et al., 2007; Fickus & Mixon, 2015; Fickus et al., 2016; Tsiligianis et al., 2017). Here we leverage the insight in (Meenakshi & Srirangarajan, 2021) to provide the analytical construction approach for generating ETFs. First calculate $\delta = \sqrt{\frac{n-r}{r(n-1)}}$ for given n and r . Let $\mathbf{S}$ denote the Seidel matrix of a strongly regular graph with n vertices as in (Holmes & Paulsen, 2004), which satisfies $\mathbf{S}_{ii} = 0, |\mathbf{S}_{ij}| = 1, \forall i \neq j$, and has eigenvalues $\frac{n-r-1}{\delta}$ and $-\frac{1}{\delta}$ with

multiplicity r and $n-r$. The Gram matrix is then defined as $\mathbf{G} = \mathbf{I}_n + \delta \mathbf{S}$. Performing the spectral decomposition $\mathbf{G} = \mathbf{U}^\top (\frac{r}{n} \mathbf{I}_r \oplus \mathbf{0}_{n-r}) \mathbf{U}$, we extract the first r rows of $\mathbf{U}$ to form the $r \times n$ frame matrix $\mathbf{X} = \sqrt{\frac{n}{r}} \mathbf{U}_{1:r}$. Finally, we can obtain the frame matrix $\mathbf{X} \in \mathbb{R}^{r \times n}$, which is an ETF.

Recent studies have shown that the gradient $\nabla f(\mathbf{W}_t)$ becomes low-rank and even rank-1 during fine-tuning (Zhao et al., 2024), and exhibits low-dimensional structures (Song et al., 2024; He et al., 2025b; Chang & Yuan, 2026). We call a matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$ to have s -effective freedom if there exists a constant s such that for every $k \in \{1, \dots, m\}$, letting $\mathcal{S}_k := \{i | z_{ki} \neq 0\}$ denote the support set of the k -th row $\mathbf{z}_k$ of $\mathbf{Z}$, then we have $\sum_{i \in \mathcal{S}_k} |z_{ki}| \leq \sqrt{s} \|\mathbf{z}_k\|_2$. Denote $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger/2} \mathbf{A}_t$, then the projection matrix can be rewritten as $\mathbf{P}_{\mathbf{A}_t} = \bar{\mathbf{A}}_t^\top \bar{\mathbf{A}}_t$. Since the property of the gradient $\nabla f(\mathbf{W}_t)$ will affect the preserved information when $\nabla f(\mathbf{W}_t)$ is unavailable, we consider the minimum proportion of gradient information retained after projection, i.e., $\Omega(\bar{\mathbf{A}}_t) = \inf_{\mathbf{W}} \frac{\|\nabla f(\mathbf{W}) \mathbf{P}_{\mathbf{A}_t}\|^2}{\|\nabla f(\mathbf{W})\|^2}$ for $\|\nabla f(\mathbf{W})\| \neq 0$. We provide the following Theorem 3.3 to derive the upper bound for the minimum proportion of preserved gradient information when the gradient $\nabla f(\mathbf{W}_t)$ is unavailable.

Theorem 3.3. For a given matrix $\mathbf{A}_t \in \mathbb{R}^{r \times n}$, denote the normalized matrix as $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger/2} \mathbf{A}_t$, then the projection matrix can be written as $\mathbf{P}_{\mathbf{A}_t} = \bar{\mathbf{A}}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger} \mathbf{A}_t = \bar{\mathbf{A}}_t^\top \bar{\mathbf{A}}_t$. For gradient $\nabla f(\mathbf{W})$ that have s -effective freedom with $s < 1 + 1/\delta$, denote $\Omega(\bar{\mathbf{A}}_t) = \inf_{\mathbf{W}} \frac{\|\nabla f(\mathbf{W}) \mathbf{P}_{\mathbf{A}_t}\|^2}{\|\nabla f(\mathbf{W})\|^2}$ for $\|\nabla f(\mathbf{W})\| \neq 0$, then the following result holds:

$$\Omega(\bar{\mathbf{A}}_t) \leq \frac{r}{n} (1 - \epsilon), \quad (18)$$

where $\epsilon = (s-1)\delta$ and $\delta = \sqrt{\frac{n-r}{r(n-1)}}$. The equality holds if and only if $\bar{\mathbf{A}}_t$ is a normalized ETF. In particular, if $\mathbf{A}_t$ is an ETF, then $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger/2} \mathbf{A}_t$ is a normalized ETF.

We leave the proof of Theorem 3.3 to Appendix A.2 due to space limitations. Theorem 3.3 reveals that if the matrix $\mathbf{A}_t$ is an ETF, then it provides the tightest guarantee for gradient information preservation. It is worth noting that if the matrix $\mathbf{A}_t$ is an ETF, then $\mathbf{A}_t \mathbf{A}_t^\top = \frac{r}{n} \mathbf{I}_r$ and consequently $\mathbf{P}_{\mathbf{A}_t} = \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger} \mathbf{A}_t = \frac{r}{n} \mathbf{A}_t^\top \mathbf{A}_t$. Therefore, Theorem 3.3 characterizes the optimality of the ETF in preserving gradient information for both LoRA update dynamics, i.e., $\mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger} \mathbf{A}_t$ in Eq. (15) and $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \frac{\alpha}{r} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top \mathbf{A}_t$ in Eq. (6) (ignoring the effect of the constant $\frac{r}{n}$ for given n and r).

3.3. Convergence Analyses for LoRA

Theorem 3.3 reveals that, at the t -th round, at least a fraction $\frac{r}{n} (1 - \epsilon)$ of the gradient information is preserved when the matrix $\mathbf{A}_t$ is an ETF. Building on this result, we next provide

the following Theorem 3.4 to establish the convergence of LoRA prototyped in Algorithm 1 for smooth loss functions.

Theorem 3.4. *Under Assumption 3.1, if the matrix $\mathbf{A}_t$ is an ETF and $\eta_{\mathbf{B}} = \frac{r}{L\alpha}$ in Algorithm 1, then:*

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla f(\mathbf{W}_t)\|^2 \leq \frac{2nL}{r(1-\epsilon)T} (f(\mathbf{W}_0) - f^*), \quad (19)$$

where $\epsilon = (s-1)\sqrt{\frac{n-r}{r(n-1)}}$, and $f^* = \inf_{\mathbf{W}} f(\mathbf{W})$.

We leave the proof of Theorem 3.4 to Appendix A.3 due to space limitations. Theorem 3.4 reveals that the averaged squared gradient norm $\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla f(\mathbf{W}_t)\|^2$ converges at a rate of $\mathcal{O}(\frac{1}{T})$ when the low-rank matrix is initialized as an ETF in Algorithm 1. When the matrix $\mathbf{A}_t$ is an ETF, then it satisfies $(\mathbf{A}_t \mathbf{A}_t^\top)^\dagger = \frac{r}{n} \mathbf{I}_r$, i.e., there is no need to calculate the inverse matrix $(\mathbf{A}_t \mathbf{A}_t^\top)^\dagger$ in Algorithm 1, thus the update rule and computational cost are the same as that in LoRA. We also consider the Polyak-Łojasiewicz (PL) condition (Polyak, 1963) as that in (Malinovsky et al., 2024), which has been widely used in gradient-based methods (Karimi et al., 2016; Haddadpour et al., 2019).

Assumption 3.5 (PL-condition). The loss function f is differentiable and satisfies the Polyak-Łojasiewicz (PL) condition, i.e., there exists a constant $\mu > 0$ such that for any $\mathbf{W} \in \mathbb{R}^{m \times n}$, the gradient satisfies:

$$\|\nabla f(\mathbf{W})\|_2^2 \geq 2\mu (f(\mathbf{W}) - f^*), \quad (20)$$

where $f^* = \inf_{\mathbf{W}} f(\mathbf{W})$.

We further provide the following Theorem 3.6 to establish the convergence of LoRA prototyped in Algorithm 1 for smooth loss functions under the PL-condition.

Theorem 3.6. *Under Assumption 3.1 and 3.5, if the matrix $\mathbf{A}_t$ is an ETF and $\eta_{\mathbf{B}} = \frac{r}{L\alpha}$ in Algorithm 1, then:*

$$f(\mathbf{W}_t) - f^* \leq \left(1 - \frac{\mu r}{nL} (1 - \epsilon)\right)^t (f(\mathbf{W}_0) - f^*), \quad (21)$$

where $\epsilon = (s-1)\sqrt{\frac{n-r}{r(n-1)}}$, and $f^* = \inf_{\mathbf{W}} f(\mathbf{W})$.

We leave the proof of Theorem 3.6 to Appendix A.4 due to space limitations. Theorem 3.6 reveals that the sequence $f(\mathbf{W}_t)$ converges to the minimizer f^* at an exponential rate when the low-rank matrix is initialized as an ETF in Algorithm 1. Note that the PL-condition is a broader concept than strong convexity; therefore, our results yield a similar convergence rate for strongly convex loss functions.

4. Discussion with Related Works

We provide Prototypical Low-Rank Adaptation as in Algorithm 1, and prove that if the low-rank matrices $\mathbf{A}_t$ or $\mathbf{B}_t$

is an ETF, then it preserves the maximum gradient information and ensures the convergence of Algorithm 1. Moreover, when $\mathbf{A}_t$ or $\mathbf{B}_t$ is an ETF, the update rule reduces to $\mathbf{B}_t \leftarrow \mathbf{B}_t^0 - \frac{r}{n} \eta_{\mathbf{B}} \nabla_{\mathbf{B}} f(\mathbf{W}_t)$ and $\mathbf{A}_t \leftarrow \mathbf{A}_t^0 - \frac{r}{n} \eta_{\mathbf{A}} \nabla_{\mathbf{A}} f(\mathbf{W}_t)$, which is consistent with the gradient-based update scheme in LoRA. In the current literature, the primary LoRA variants, e.g., LoRA (Hu et al., 2022), FLORA (Hao et al., 2024), AsymLoRA (Zhu et al., 2024), ReLoRA (Lialin et al., 2024), RAC-LoRA (Malinovsky et al., 2024), typically initialize the low-rank matrices with Gaussian distributions. Therefore, the Gaussian initialization in these methods can be directly replaced with ETF initialization without introducing additional computational overhead. Theoretically, ETF initialization preserves more gradient information than Gaussian initialization, thus facilitating effective gradient descent and enhancing the performance in primary LoRA variants, which have been validated in Section 5.2 and 5.3.

The term $(\mathbf{A}_t \mathbf{A}_t^\top)^\dagger$ has also appeared in the update rule in some works (Wang et al., 2025b; Yu et al., 2025; Zhang & Pilanci, 2024), which heuristically gave the update rule. Instead, we update the low-rank matrices based on Eq. (14) and Eq. (16) to ensure the descent property of the loss function, i.e., $f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t)$. Malinovsky et al. (2024) gave a similar form of Eq. (14), but failed to provide the update rule to obtain this form. Malinovsky et al. (2024) also provided theoretical analyses with the assumption that the smallest singular value of the induced projection matrix is positive, which is impractical since it implies that the rank of the LoRA matrices must be equal to the dimension of the pre-trained weights. Moreover, the above works do not analyze the properties of the LoRA matrices in preserving the gradient information, while we prove that the ETFs can achieve the maximum lower bound for gradient information preservation, and further provide convergence analyses for LoRA with ETF initialization.

Recent studies utilized the Gaussian initialization (Hu et al., 2022) and the Stiefel manifold (He et al., 2025b; Hao et al., 2024) to capture the gradient information. However, generic Stiefel manifold elements cannot effectively preserve the gradient information. Consider two common examples that satisfy the Stiefel manifold: **I.** Suppose the low rank matrix $\mathbf{Q}$ consists of r rows of an $n \times n$ orthogonal matrix with $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_r$, thus $\mathbf{Q}$ belongs to the Stiefel manifold. Note that $\mathbf{Q}^\top \mathbf{Q} = \mathbf{U}^\top (\mathbf{I}_r \oplus \mathbf{0}_{n-r}) \mathbf{U}$, where $\mathbf{U}$ is an orthogonal matrix. If the non-zero elements of $\nabla f(\mathbf{W})$ concentrate in the last $n-r$ coordinates, then $\|\nabla f(\mathbf{W}) \mathbf{P}_{\mathbf{Q}}\| = 0$, where $\mathbf{P}_{\mathbf{Q}} = \mathbf{Q}^\top (\mathbf{Q}\mathbf{Q}^\top)^\dagger \mathbf{Q} = \mathbf{Q}^\top \mathbf{Q}$, indicating complete energy loss. **II.** Consider the low rank matrix $\mathbf{Q}$ with entries i.i.d. as normal $\mathcal{N}(0, 1)$, then $\mathbb{E}[\mathbf{Q}\mathbf{Q}^\top] = \mathbf{I}_r$, thus $\mathbf{Q}$ belongs to the Stiefel manifold (He et al., 2025b). Note that $\mathbf{Q}^\top \mathbf{Q} = \mathbf{V}^\top \mathbf{V}$, where $\mathbf{V}$ is an equivalence class of row-orthonormal matrices, reducing this to the first example. Therefore, if $\mathbf{A}_t$ is initialized as Gaussian or a general

Stiefel manifold, the proportion of information preserved depends on the alignment between $\mathbf{A}_t$ and the gradient directions. In contrast, if $\mathbf{A}_t$ is an ETF, then it can effectively project the gradient information onto the subspace induced by $\mathbf{A}_t$, since we have proven that the ETFs can achieve the maximum lower bound for preserving gradient information.

5. Experiments

We have theoretically revealed that the ETF structure can achieve the maximum lower bound for gradient information preservation when the gradient $\nabla f(\mathbf{W}_t)$ is unavailable, which suggests that the ETF structure is a preferable choice for initializing the low-rank matrices. Thus, we visualize the effectiveness of ETF in preserving gradient information in Section 5.1, and conduct the experiments on natural language understanding tasks in Section 5.2 and natural language generation tasks in Section 5.3 to validate this.

5.1. Visualization of the Effectiveness of ETF

To visualize the effects of ETF initializations in preserving energy, we generate a random $n \times n$ matrix as $\nabla f(\mathbf{W})$ where $n = 100$. For a rank- r matrix $\mathbf{A}$ where $r = 4$, we calculate $\frac{\|\nabla f(\mathbf{W})\mathbf{P}_\mathbf{A}\|^2}{\|\nabla f(\mathbf{W})\|^2}$ to measure the proportion of preserved energy. We initialize $\mathbf{A}$ from three different approaches: ETF, Gaussian, and orthogonal. Specifically, the orthogonal initialization implies that $\mathbf{A}$ is formed by selecting r rows from an $n \times n$ orthogonal matrix, *i.e.*, $\mathbf{A}$ belongs to the Stiefel manifold. We conduct 100 independent trials and depict the results in Figure 1. The results in Figure 1 show that ETF initialization consistently maintains higher preservation ratios than Gaussian and Orthogonal initialization on both average and lower bound. This result validates our theoretical analyses that ETF can provide a more effective subspace for gradient preservation than Gaussian and general Stiefel manifold elements.

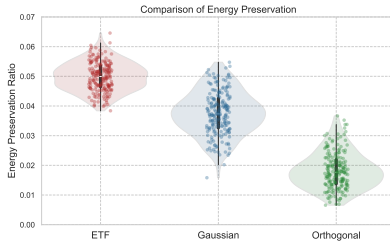


Figure 1. Comparison of energy preservation across ETF, Gaussian and Orthogonal initialization.

5.2. Natural Language Understanding Tasks

Methods. We compare the following LoRA variants with and without ETF initialization to validate the effectiveness of ETF: **I.** Standard LoRA and its variants, *i.e.*, LoRA (Hu

et al., 2022), LoRA-GA (Wang et al., 2024) and PiSSA (Meng et al., 2024); **II.** Fix-A LoRA variants, *i.e.*, FLORA (Hao et al., 2024) and AsymLoRA (Zhu et al., 2024); **III.** Periodically optimized LoRA variants, *i.e.*, ReLoRA (Lialin et al., 2024); **IV.** Fix-A LoRA variants with periodic optimization, *i.e.*, RAC-LoRA (Malinovsky et al., 2024). The approach for constructing ETFs in the experiments is provided in Appendix B.1 due to space limitations. The notation w/o ETF and w/ ETF refer to these methods with their default initialization and ETF initialization, respectively.

Setup. We evaluate these methods on several GLUE benchmark datasets (Wang et al., 2019), *i.e.*, CoLA (Warstadt et al., 2019), SST-2 (Socher et al., 2013), MRPC (Dolan & Brockett, 2005), QQP¹, MNLI (Williams et al., 2018), QNLI (Rajpurkar et al., 2016), and RTE (Wang et al., 2019). We use the RoBERTa (Liu et al., 2019) model and the Llama-3.1-8B (Dubey et al., 2024) model as the backbone. The detailed setups are provided in Appendix B.2.

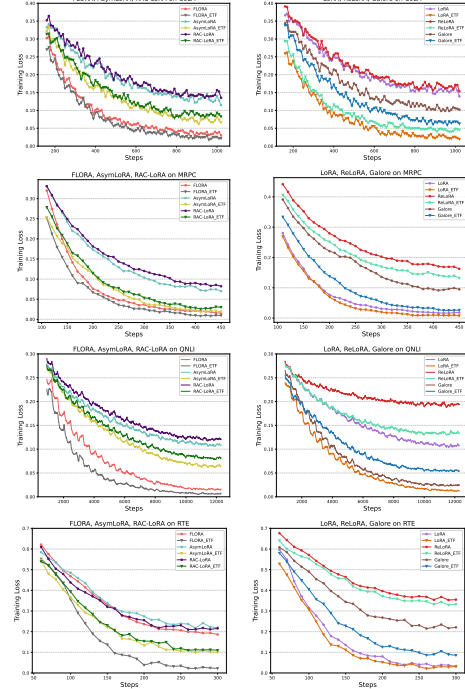


Figure 2. Convergence performance of LoRA variants with and without ETF, where the rank is 8.

Result Analysis. We repeat all the experiments over 5 times and report the average accuracy with standard deviation of the RoBERTa and Llama-3.1-8B model for rank 8 in Table 1 and Table 7 in Appendix B.2, respectively. Results of these methods of the RoBERTa and Llama-3.1-8B model for rank 4 and 16 are shown in Table 4, Table 5, and Table 6, Table 8 respectively in Appendix B.2. The results show that the LoRA variants with ETF consistently outperform their

¹data.quora.com/First-Quora-Dataset-Release-Question-Pairs

Table 1. Test accuracy(%) of RoBERTa with LoRA variants (with/without ETF) on GLUE benchmark datasets (rank 8).

Method	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
LoRA	w/o	57.95 $\pm$ 1.23	94.19 $\pm$ 0.05	87.83 $\pm$ 0.99	90.38 $\pm$ 0.04	86.27 $\pm$ 0.06	91.53 $\pm$ 0.05	73.05 $\pm$ 1.47
	w/	62.18 $\pm$ 0.87 (+4.23)	95.22 $\pm$ 0.11 (+1.03)	88.89 $\pm$ 1.29 (+1.06)	91.47 $\pm$ 0.06 (+1.09)	87.48 $\pm$ 0.08 (+1.21)	92.65 $\pm$ 0.11 (+1.12)	74.97 $\pm$ 1.73 (+1.92)
LoRA-GA	w/o	58.70 $\pm$ 0.82	94.43 $\pm$ 0.23	87.52 $\pm$ 0.62	90.45 $\pm$ 0.02	86.37 $\pm$ 0.16	91.88 $\pm$ 0.12	73.48 $\pm$ 1.81
	w/	62.18 $\pm$ 0.87 (+3.48)	95.22 $\pm$ 0.11 (+0.79)	88.89 $\pm$ 1.29 (+1.37)	91.47 $\pm$ 0.06 (+1.02)	87.48 $\pm$ 0.08 (+1.11)	92.65 $\pm$ 0.11 (+0.77)	74.97 $\pm$ 1.73 (+1.49)
PiSSA	w/o	58.14 $\pm$ 0.71	94.31 $\pm$ 0.19	87.49 $\pm$ 1.26	90.32 $\pm$ 0.02	85.94 $\pm$ 0.05	91.56 $\pm$ 0.14	74.37 $\pm$ 1.64
	w/	62.18 $\pm$ 0.87 (+4.04)	95.22 $\pm$ 0.11 (+0.91)	88.89 $\pm$ 1.29 (+1.40)	91.47 $\pm$ 0.06 (+1.15)	87.48 $\pm$ 0.08 (+1.54)	92.65 $\pm$ 0.11 (+1.09)	74.97 $\pm$ 1.73 (+0.60)
FLORA	w/o	59.87 $\pm$ 1.27	93.96 $\pm$ 0.30	87.01 $\pm$ 0.53	90.29 $\pm$ 0.06	86.51 $\pm$ 0.04	91.92 $\pm$ 0.15	72.08 $\pm$ 0.55
	w/	62.42 $\pm$ 1.18 (+2.55)	95.22 $\pm$ 0.11 (+1.26)	88.89 $\pm$ 0.64 (+1.88)	91.37 $\pm$ 0.04 (+1.08)	87.52 $\pm$ 0.06 (+1.01)	92.94 $\pm$ 0.08 (+1.02)	74.85 $\pm$ 0.90 (+2.77)
AsymLoRA	w/o	58.47 $\pm$ 0.67	94.15 $\pm$ 0.12	87.52 $\pm$ 0.75	90.03 $\pm$ 0.18	85.86 $\pm$ 0.13	91.43 $\pm$ 0.14	72.20 $\pm$ 1.28
	w/	60.57 $\pm$ 0.54 (+2.10)	95.11 $\pm$ 0.14 (+0.96)	88.64 $\pm$ 0.42 (+1.12)	91.05 $\pm$ 0.05 (+1.02)	86.96 $\pm$ 0.10 (+1.10)	92.50 $\pm$ 0.11 (+1.07)	73.65 $\pm$ 1.56 (+1.45)
ReLoRA	w/o	57.14 $\pm$ 1.94	91.97 $\pm$ 0.34	87.01 $\pm$ 0.87	89.66 $\pm$ 0.02	82.79 $\pm$ 0.62	90.56 $\pm$ 0.26	71.24 $\pm$ 1.11
	w/	62.11 $\pm$ 0.90 (+4.97)	93.35 $\pm$ 0.23 (+1.38)	88.07 $\pm$ 0.90 (+1.06)	90.70 $\pm$ 0.02 (+1.04)	85.16 $\pm$ 0.02 (+2.37)	91.97 $\pm$ 0.10 (+1.41)	72.44 $\pm$ 0.90 (+1.20)
RAC-LoRA	w/o	55.35 $\pm$ 1.23	93.69 $\pm$ 0.30	86.93 $\pm$ 0.37	90.02 $\pm$ 0.05	85.89 $\pm$ 0.01	91.48 $\pm$ 0.10	71.84 $\pm$ 1.64
	w/	58.84 $\pm$ 0.87 (+3.49)	94.72 $\pm$ 0.19 (+1.03)	88.56 $\pm$ 0.42 (+1.63)	91.01 $\pm$ 0.02 (+0.99)	86.92 $\pm$ 0.11 (+1.03)	92.49 $\pm$ 0.13 (+1.01)	73.65 $\pm$ 1.35 (+1.81)

Table 2. Test accuracy(%) of RoBERTa with Galore (with/without ETF) on GLUE benchmark datasets.

Rank	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
4	w/o	58.46 $\pm$ 0.52	95.03 $\pm$ 0.05	86.85 $\pm$ 0.61	91.08 $\pm$ 0.04	87.32 $\pm$ 0.11	92.64 $\pm$ 0.09	68.83 $\pm$ 1.33
	w/	59.55 $\pm$ 0.48 (+1.09)	94.88 $\pm$ 0.26 (-0.15)	88.32 $\pm$ 0.95 (+1.47)	90.15 $\pm$ 0.04 (-0.93)	87.18 $\pm$ 0.08 (-0.14)	92.35 $\pm$ 0.19 (-0.29)	71.72 $\pm$ 1.12 (+2.89)
8	w/o	59.05 $\pm$ 0.94	95.15 $\pm$ 0.22	87.09 $\pm$ 0.12	91.25 $\pm$ 0.06	87.47 $\pm$ 0.02	92.92 $\pm$ 0.07	71.84 $\pm$ 1.84
	w/	61.61 $\pm$ 0.16 (+2.56)	95.12 $\pm$ 0.06 (-0.03)	88.48 $\pm$ 0.20 (+1.39)	91.29 $\pm$ 0.04 (+0.04)	87.66 $\pm$ 0.03 (+0.19)	92.58 $\pm$ 0.17 (-0.34)	74.01 $\pm$ 0.78 (+2.17)
16	w/o	59.91 $\pm$ 1.00	95.22 $\pm$ 0.18	87.13 $\pm$ 0.86	91.42 $\pm$ 0.04	87.64 $\pm$ 0.10	92.71 $\pm$ 0.12	73.65 $\pm$ 2.21
	w/	61.76 $\pm$ 0.67 (+1.85)	95.26 $\pm$ 0.13 (+0.04)	88.40 $\pm$ 1.10 (+1.27)	91.45 $\pm$ 0.13 (+0.03)	87.67 $\pm$ 0.03 (+0.03)	92.95 $\pm$ 0.06 (+0.24)	74.13 $\pm$ 1.33 (+0.48)

default initialization counterparts across all datasets. This result aligns with our Theorem 3.3, which states that the ETF structure provides the maximum lower bound for preserving gradient energy since it can effectively map the information from each row of the gradient $\nabla f(\mathbf{W})$ onto the subspace induced by $\mathbf{A}$. In contrast, existing works have revealed that Gaussian initialization, as in LoRA, FLORA, AsymLoRA, ReLoRA, and RAC-LoRA, leads to uninformative gradients for a long time and slow convergence since the subspace induced by $\mathbf{A}$ fails to preserve sufficient gradient information (Meng et al., 2024). Although LoRA-GA and PiSSA perform SVD on the gradient and pre-trained model weights respectively for their initialization, their performance is inferior to our ETF initialization since estimating the gradients using stochastic gradients from a limited number of batches introduces bias, and decomposing the parameter matrix may not align with gradient-based updates. Therefore, these results imply that ETF can facilitate effective gradient descent and enhance the fine-tuning performance of LoRA.

Comparison with GaLore. To further validate the effectiveness of ETF in preserving gradient information, we compare ETF with the method that continually utilizes the gradient information in the fine-tuning process, *i.e.*, GaLore (Zhao et al., 2024), which requires access to the gradient $\nabla f(\mathbf{W})$ and performs SVD on $\nabla f(\mathbf{W})$ to initialize the low-rank matrices. For comparison, we use ETF to initialize the low-rank matrices in GaLore, which does not require the gradient

$\nabla f(\mathbf{W})$. Comparisons of GaLore with and without ETF of RoBERTa and Llama-3.1-8B for rank 4, 8, and 16 are shown in Table 2 and Table 9 in Appendix B.2, respectively. The results show that ETF-initialized GaLore achieves comparable performance to its default SVD initialization. The potential reason lies in that GaLore recomputes the low-rank matrices with SVD every fixed number of iterations since performing SVD is computationally expensive, *i.e.*, it uses the current gradient to estimate future gradients. In contrast, without access to the gradient, ETF universally preserves effective gradient information across varying gradients. The results validate the effectiveness of ETF in ensuring robust and efficient gradient descent by maximizing the lower bound of gradient information preservation as in Theorem 3.3.

Convergence Performance. As revealed in Theorem 3.3, the ETF structure provides the maximum lower bound for preserving gradient energy and can facilitate effective gradient descent. To further validate this, we depict the convergence curves of these LoRA variants with and without ETF of the RoBERTa model for rank 8 on some representative datasets in Figure 2 due to space limitations. Note that we omit the convergence curves of LoRA-GA and PiSSA, as they mainly modify the initialization approach of LoRA. Therefore, their ETF-initialization versions exhibit the same convergence performance as the ETF-initialization version of LoRA. The complete convergence curves for these methods of the RoBERTa model for rank 4, 8, 16 are shown in

Table 3. Test score(%) of Llama-3.1-8B and Mistral-7B-v0.1 with LoRA variants (with/without ETF) on NLG datasets (rank 8).

Method	ETF	Llama-3.1-8B				Mistral-7B-v0.1			
		GSM8K	HumanEval	MBPP	MT-Bench	GSM8K	HumanEval	MBPP	MT-Bench
LoRA	w/o	68.33 $\pm$ 0.27	42.48 $\pm$ 0.58	50.23 $\pm$ 0.34	6.06 $\pm$ 0.04	69.61 $\pm$ 0.19	29.26 $\pm$ 1.36	40.21 $\pm$ 0.16	6.13 $\pm$ 0.03
	w/	71.07 $\pm$ 0.19 (+2.74)	46.34 $\pm$ 1.22 (+3.86)	51.45 $\pm$ 0.57 (+1.22)	6.45 $\pm$ 0.11 (+0.39)	71.40 $\pm$ 0.36 (+1.79)	32.31 $\pm$ 0.53 (+3.05)	42.07 $\pm$ 0.58 (+1.86)	6.53 $\pm$ 0.06 (+0.40)
LoRA-GA	w/o	70.94 $\pm$ 0.12	43.39 $\pm$ 0.66	49.01 $\pm$ 0.12	6.32 $\pm$ 0.05	70.07 $\pm$ 0.27	31.09 $\pm$ 0.75	41.35 $\pm$ 0.16	6.06 $\pm$ 0.03
	w/	71.07 $\pm$ 0.19 (+0.13)	46.34 $\pm$ 1.22 (+2.95)	51.45 $\pm$ 0.57 (+2.44)	6.45 $\pm$ 0.11 (+0.13)	71.40 $\pm$ 0.36 (+1.33)	32.31 $\pm$ 0.53 (+1.22)	42.07 $\pm$ 0.58 (+0.72)	6.53 $\pm$ 0.06 (+0.47)
PiSSA	w/o	67.08 $\pm$ 0.24	42.31 $\pm$ 0.73	49.84 $\pm$ 0.48	6.37 $\pm$ 0.09	70.96 $\pm$ 0.91	30.87 $\pm$ 0.25	40.24 $\pm$ 0.66	6.15 $\pm$ 0.04
	w/	71.07 $\pm$ 0.19 (+3.99)	46.34 $\pm$ 1.22 (+4.03)	51.45 $\pm$ 0.57 (+1.61)	6.45 $\pm$ 0.11 (+0.08)	71.40 $\pm$ 0.36 (+0.44)	32.31 $\pm$ 0.53 (+1.44)	42.07 $\pm$ 0.58 (+1.83)	6.53 $\pm$ 0.06 (+0.38)
FLORA	w/o	71.22 $\pm$ 0.12	43.90 $\pm$ 0.75	46.15 $\pm$ 0.31	6.13 $\pm$ 0.06	70.34 $\pm$ 1.37	38.31 $\pm$ 1.33	40.83 $\pm$ 0.29	5.96 $\pm$ 0.07
	w/	73.40 $\pm$ 0.04 (+2.18)	46.95 $\pm$ 1.22 (+3.05)	48.67 $\pm$ 0.21 (+2.52)	6.26 $\pm$ 0.03 (+0.13)	71.94 $\pm$ 0.57 (+1.60)	39.63 $\pm$ 0.85 (+1.32)	42.52 $\pm$ 0.14 (+1.69)	6.07 $\pm$ 0.05 (+0.11)
AsymLoRA	w/o	63.03 $\pm$ 0.35	42.07 $\pm$ 0.86	50.07 $\pm$ 0.41	5.91 $\pm$ 0.05	64.93 $\pm$ 0.32	26.37 $\pm$ 1.06	39.63 $\pm$ 0.45	6.21 $\pm$ 0.03
	w/	66.06 $\pm$ 0.41 (+3.03)	43.59 $\pm$ 0.30 (+1.52)	51.22 $\pm$ 0.19 (+1.15)	6.25 $\pm$ 0.10 (+0.34)	66.84 $\pm$ 0.50 (+1.91)	29.87 $\pm$ 1.22 (+3.50)	40.62 $\pm$ 0.09 (+0.99)	6.26 $\pm$ 0.02 (+0.05)
ReLoRA	w/o	67.25 $\pm$ 0.30	43.29 $\pm$ 2.17	50.53 $\pm$ 0.33	6.38 $\pm$ 0.06	67.60 $\pm$ 0.55	29.41 $\pm$ 0.67	40.05 $\pm$ 0.51	6.03 $\pm$ 0.01
	w/	70.35 $\pm$ 0.30 (+3.10)	46.95 $\pm$ 0.61 (+3.66)	51.04 $\pm$ 0.52 (+0.51)	6.42 $\pm$ 0.07 (+0.04)	70.88 $\pm$ 0.38 (+3.28)	30.48 $\pm$ 0.56 (+1.07)	41.03 $\pm$ 0.05 (+0.98)	6.38 $\pm$ 0.02 (+0.35)
RAC-LoRA	w/o	60.30 $\pm$ 0.27	41.16 $\pm$ 0.91	50.45 $\pm$ 0.46	6.02 $\pm$ 0.01	64.52 $\pm$ 0.90	27.89 $\pm$ 0.52	40.28 $\pm$ 0.36	5.93 $\pm$ 0.09
	w/	63.33 $\pm$ 0.14 (+3.03)	43.44 $\pm$ 1.80 (+2.28)	52.26 $\pm$ 0.30 (+1.81)	6.07 $\pm$ 0.08 (+0.05)	66.61 $\pm$ 0.71 (+2.09)	29.26 $\pm$ 0.44 (+1.37)	41.47 $\pm$ 0.16 (+1.19)	6.16 $\pm$ 0.07 (+0.23)
GaLore	w/o	76.21 $\pm$ 0.25	49.26 $\pm$ 0.30	50.27 $\pm$ 0.13	6.40 $\pm$ 0.02	71.53 $\pm$ 1.28	43.46 $\pm$ 1.74	46.35 $\pm$ 0.25	6.20 $\pm$ 0.01
	w/	78.71 $\pm$ 0.75 (+2.50)	52.43 $\pm$ 0.61 (+3.17)	50.42 $\pm$ 0.42 (+0.15)	6.54 $\pm$ 0.07 (+0.14)	72.93 $\pm$ 0.37 (+1.40)	46.95 $\pm$ 1.22 (+3.49)	48.18 $\pm$ 0.41 (+1.83)	6.43 $\pm$ 0.04 (+0.23)

Figure 3, 4, 5 in Appendix B.2. The convergence results show that these methods with ETF initialization converge faster than their default initialization counterparts. In addition, GaLore with ETF also achieves comparable convergence speed to the original GaLore with SVD initialization, which validates the effectiveness of ETF in facilitating gradient descent as revealed in Theorem 3.3.

Robustness to Model and Rank. The results of varying models (RoBERTa and Llama-3.1-8B) for varying ranks (4, 8, 16) show that these LoRA variants with ETF initialization consistently achieve superior performance and convergence across all models and ranks, confirming the robustness of ETF to varying models and ranks in preserving gradient information and accelerating convergence.

5.3. Natural Language Generation Tasks

We further evaluate these methods on GSM8K (Cobbe et al., 2021), HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021), and MT-Bench (Zheng et al., 2024a) benchmarks, including math reasoning, code generation, and instruction following tasks. We use the Llama-3.1-8B (Dubey et al., 2024) and Mistral-7B-v0.1 (Jiang et al., 2023) model implemented by the HuggingFace Transformers library (Wolf et al., 2020) as the backbone. The detailed setups are provided in Appendix B.3. We repeat all the experiments over 5 times and report the average score with standard deviation of the Llama-3.1-8B and Mistral-7B-v0.1 model for rank 8 in Table 3. We also show the corresponding results for rank 4 and 16 in Table 10 and Table 11 in Appendix B.3. The results show that these LoRA variants with ETF consistently outperform their default initialization counterparts on natural language generation tasks. This performance improvement also extends to GaLore, further supporting that the ETF structure provides the maximum lower bound

for preserving gradient energy, further facilitates effective gradient descent and enhances the fine-tuning performance as revealed in Theorem 3.3. The results of varying models (Llama-3.1-8B, Mistral-7B-v0.1) for varying ranks (4, 8, 16) further confirm the robustness of ETF to varying models and ranks in preserving gradient information and accelerating convergence on natural language generation tasks.

6. Conclusion

In this paper, we provide a theoretical understanding of the update dynamics of LoRA. We reveal that the update dynamics can be viewed as projecting the gradient $\nabla f(\mathbf{W})$ onto the subspace spanned by $\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^\dagger\mathbf{A}$, and prove that when the gradient $\nabla f(\mathbf{W})$ is unavailable, if $\mathbf{A}$ is an Equiangular Tight Frame (ETF), $\mathbf{A}^\top\mathbf{A}$ and $\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^\dagger\mathbf{A}$ can preserve the maximum information of the gradient $\nabla f(\mathbf{W})$. Thus, initializing $\mathbf{A}$ as an ETF is the optimal solution for low-rank adaptation when the gradient $\nabla f(\mathbf{W})$ is unavailable. Furthermore, we establish the convergence of Low-Rank Adaptation with a rate of $\mathcal{O}(\frac{1}{T})$ when $\mathbf{A}$ is an ETF. Experiments show that initializing the low-rank matrices as ETFs significantly outperforms the commonly used Gaussian initialization for existing primary LoRA variants.

Acknowledgements

This work was supported by the National Key R&D Program of China (2024YFC3808203), Basic Research Program of Jiangsu (BK20253021), National Science Foundation of China (62276125), Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (JYB2025XDXM118), “111 Center” (B26023), and Collaborative Innovation Center of Novel Software Technology and Industrialization.

Impact Statement

This work provides a theoretical understanding of the update dynamics of Low-Rank Adaptation (LoRA), establishes Equiangular Tight Frame (ETF) as an effective initialization approach for LoRA, and addresses the critical issue of the convergence in LoRA. We believe that our work provides a theoretical foundation for this line of methods and will provide novel insights for future theoretical research and algorithmic development. Our work will contribute positively to the advancement of the machine learning community and entails no adverse societal impacts.

References

- Aghajanyan, A., Gupta, S., and Zettlemoyer, L. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL/IJCNLP’21)*, pp. 7319–7328, 2021.
- Austin, J., Odena, A., Nye, M. I., Bosma, M., Michalewski, H., Dohan, D., Jiang, E., Cai, C. J., Terry, M., Le, Q. V., and Sutton, C. Program synthesis with large language models. *arXiv:2108.07732*, 2021.
- Bottou, L., Curtis, F. E., and Nocedal, J. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, 2018.
- Boyd, S. and Vandenberghe, L. *Convex optimization*. Cambridge university press, 2004.
- Chang, D. and Yuan, G. MGUP: A momentum-gradient alignment update policy for stochastic optimization. In *Advances in Neural Information Processing Systems 39 (NeurIPS’26)*, 2026.
- Chen, M., Tworek, J., Jun, H., Yuan, Q., de Oliveira Pinto, H. P., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., Puri, R., Krueger, G., Petrov, M., Khlaaf, H., and et al. Evaluating large language models trained on code. *arxiv:2107.03374*, 2021.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arxiv:2110.14168*, 2021.
- Comanici, G., Bieber, E., Schaekermann, M., Pasupat, I., Sachdeva, N., Dhillon, I., Blistein, M., Ram, O., Zhang, D., Rosen, E., Marris, L., and et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025.
- DeepSeek-AI, Liu, A., Feng, B., Xue, B., Wang, B., Wu, B., Lu, C., Zhao, C., Deng, C., Zhang, C., Ruan, C., Dai, D., Guo, D., Yang, D., Chen, D., Ji, D., Li, E., and et al. Deepseek-v3 technical report, 2025.
- Ding, N., Lv, X., Wang, Q., Chen, Y., Zhou, B., Liu, Z., and Sun, M. Sparse low-rank adaptation of pre-trained language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP’23)*, pp. 4133–4145, 2023.
- Dolan, W. B. and Brockett, C. Automatically constructing a corpus of sentential paraphrases. In *Proceedings of the 3rd International Workshop on Paraphrasing (IWP@IJCNLP’05)*, 2005.
- Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Yang, A., Fan, A., Goyal, A., Hartshorn, A., Yang, A., Mitra, A., Sravankumar, A., Korenev, A., Hinsvark, A., and et al. The llama 3 herd of models. *arxiv:2407.21783*, 2024.
- Fickus, M., Mixon, D. G., and Jasper, J. Equiangular tight frames from hyperovals. *IEEE Transactions on Information Theory*, 62(9):5225–5236, 2016.
- Fickus, M. C. and Mixon, D. G. Tables of the existence of equiangular tight frames. *arXiv:1504.00253*, 2015.
- Guo, P., Zeng, S., Wang, Y., Fan, H., Wang, F., and Qu, L. Selective aggregation for low-rank adaptation in federated learning. In *Proceedings of the 13th International Conference on Learning Representations (ICLR’25)*, 2025.
- Haddadpour, F., Kamani, M. M., Mahdavi, M., and Cadambe, V. R. Local sgd with periodic averaging: Tighter analysis and adaptive synchronization. In *Advances in Neural Information Processing Systems 32 (NeurIPS’19)*, pp. 1108–11092, 2019.
- Hao, Y., Cao, Y., and Mou, L. Flora: Low-rank adapters are secretly gradient compressors. In *Proceedings of the 41st International Conference on Machine Learning (ICML’24)*, 2024.
- Hayou, S., Ghosh, N., and Yu, B. Lora+: Efficient low rank adaptation of large models. In *Proceedings of the 41st International Conference on Machine Learning (ICML’24)*, 2024.
- He, H., Ye, P., Ren, Y., Yuan, Y., and Chen, L. Gora: Gradient-driven adaptive low rank adaptation. In *Advances in Neural Information Processing Systems 38 (NeurIPS’25)*, 2025a.
- He, Y., Li, P., Hu, Y., Chen, C., and Yuan, K. Subspace optimization for large language models with convergence guarantees. In *Proceedings of the 42nd International*

- Conference on Machine Learning (ICML'25). OpenReview.net, 2025b.
- Holmes, R. B. and Paulsen, V. I. Optimal frames for erasures. *Linear Algebra and its Applications*, 377:31–51, 2004.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. Lora: Low-rank adaptation of large language models. In *Proceedings of the 10th International Conference on Learning Representations (ICLR'22)*, 2022.
- Isaacs, J. C. Stochastic constructions of equiangular tight frames. In *2011 Digital Signal Processing and Signal Processing Education Meeting (DSP/SPE)*, pp. 66–71, 2011.
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de Las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M., Stock, P., Scao, T. L., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mistral 7b. *arXiv:2310.06825*, 2023.
- Kalajdziewski, D. A rank stabilization scaling factor for fine-tuning with lora. *arXiv:2312.03732*, 2023.
- Karimi, H., Nutini, J., and Schmidt, M. Linear convergence of gradient and proximal-gradient methods under the polyak-łojasiewicz condition. In *Proceedings of the 2016 European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD'16)*, volume 9851, pp. 795–811, 2016.
- Khaled, A. and Richtárik, P. Better theory for SGD in the nonconvex world. *Transactions on Machine Learning Research*, 2023, 2023.
- Kim, J., Kim, J., and Ryu, E. K. Lora training provably converges to a low-rank global minimum or it fails loudly (but it probably won't fail). In *Proceedings of the 42st International Conference on Machine Learning (ICML'25)*, 2025.
- Kopiczko, D. J., Blankevoort, T., and Asano, Y. M. Vera: Vector-based random matrix adaptation. In *Proceedings of the 12th International Conference on Learning Representations (ICLR'24)*, 2024.
- Li, C., Farkhoor, H., Liu, R., and Yosinski, J. Measuring the intrinsic dimension of objective landscapes. In *Proceedings of the 6th International Conference on Learning Representations (ICLR'18)*, 2018.
- Lialin, V., Muckatira, S., Shivagunde, N., and Rumshisky, A. Relora: High-rank training through low-rank updates. In *Proceedings of the 12th International Conference on Learning Representations (ICLR'24)*, 2024.
- Liu, S., Wang, C., Yin, H., Molchanov, P., Wang, Y. F., Cheng, K., and Chen, M. Dora: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2024.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., and Stoyanov, V. Roberta: A robustly optimized BERT pretraining approach. *arXiv:1907.11692*, 2019.
- Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. In *Proceedings of the 7th International Conference on Learning Representations (ICLR'19)*, 2019.
- Malinovsky, G., Michieli, U., Hammoud, H. A. A. K., Ceritli, T., Elesedy, H., Ozay, M., and Richtárik, P. Randomized asymmetric chain of lora: The first meaningful theoretical framework for low-rank adaptation. *arxiv:2410.08305*, 2024.
- Meenakshi and Srirangarajan, S. Incoherent and robust projection matrix design based on equiangular tight frame. *IEEE Access*, 9:131462–131475, 2021.
- Meng, F., Wang, Z., and Zhang, M. Pissa: Principal singular values and singular vectors adaptation of large language models. In Globersons, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J. M., and Zhang, C. (eds.), *Advances in Neural Information Processing Systems 37 (NeurIPS'24)*, 2024.
- Nesterov, Y. et al. *Lectures on convex optimization*, volume 137. Springer, 2018.
- Nesterov, Y. E. *Introductory Lectures on Convex Optimization - A Basic Course*. Springer, 2004.
- OpenAI. GPT-4 technical report. *arXiv:2303.08774*, 2023.
- Polyak, B. Gradient methods for the minimisation of functionals. *USSR Computational Mathematics and Mathematical Physics*, 3(4):864–878, 1963.
- Rajpurkar, P., Zhang, J., Lopyrev, K., and Liang, P. Squad: 100, 000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP'16)*, pp. 2383–2392, 2016.
- Ren, P., Shi, C., Wu, S., Zhang, M., Ren, Z., de Rijke, M., Chen, Z., and Pei, J. MELoRA: Mini-ensemble low-rank adapters for parameter-efficient fine-tuning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL'24)*, pp. 3052–3064, 2024.
- Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C. D., Ng, A. Y., and Potts, C. Recursive deep models

- for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP'13)*, pp. 1631–1642, 2013.
- Song, W., Li, Z., Zhang, L., hai zhao, and Du, B. Sparse is enough in fine-tuning pre-trained large language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2024.
- Strohmer, T. and Heath, R. W. Grassmannian frames with applications to coding and communication. *Applied and Computational Harmonic Analysis*, 14(3):257–275, 2003.
- Sun, Y., Li, Z., Li, Y., and Ding, B. Improving lora in privacy-preserving federated learning. In *Proceedings of the 12th International Conference on Learning Representations (ICLR'24)*, 2024.
- Sustik, M. A., Tropp, J. A., Dhillon, I. S., and Heath Jr, R. W. On the existence of equiangular tight frames. *Linear Algebra and its applications*, 426(2-3):619–635, 2007.
- Tahir, B., Schwarz, S., and Rupp, M. Constructing grassmannian frames by an iterative collision-based packing. *IEEE Signal Processing Letters*, 26(7):1056–1060, 2019.
- Tian, C., Shi, Z., Guo, Z., Li, L., and zhong Xu, C. HydraloRA: An asymmetric loRA architecture for efficient fine-tuning. In *Advances in Neural Information Processing Systems 37 (NeurIPS'24)*, 2024.
- Tsiligianni, E., Kondi, L. P., and Katsaggelos, A. K. Approximate equiangular tight frames for compressed sensing and CDMA applications. *EURASIP Journal on Advances in Signal Processing*, 2017:66, 2017.
- Valipour, M., Rezagholizadeh, M., Kobzyev, I., and Ghodsi, A. Dylora: Parameter-efficient tuning of pre-trained models using dynamic search-free low-rank adaptation. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL'23)*, pp. 3266–3279, 2023.
- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., and Bowman, S. R. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 7th International Conference on Learning Representations (ICLR'19)*, 2019.
- Wang, H., Li, Y., Wang, S., Chen, G., and Chen, Y. MiLoRA: Harnessing minor singular components for parameter-efficient LLM finetuning. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL'25)*, pp. 4823–4836, 2025a.
- Wang, S., Yu, L., and Li, J. Lora-ga: Low-rank adaptation with gradient approximation. In *Advances in Neural Information Processing Systems 37 (NeurIPS'24)*, 2024.
- Wang, Z., Liang, J., He, R., Wang, Z., and Tan, T. Lora-pro: Are low-rank adapters properly optimized? In *Proceedings of the 13th International Conference on Learning Representations (ICLR'25)*, 2025b.
- Warstadt, A., Singh, A., and Bowman, S. R. Neural network acceptability judgments. *Transactions of the Association for Computational Linguistics*, 7:625–641, 2019.
- Welch, L. R. Lower bounds on the maximum cross correlation of signals (corresp.). *IEEE Transactions on Information Theory*, 20(3):397–399, 1974.
- Williams, A., Nangia, N., and Bowman, S. R. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT'18)*, pp. 1112–1122, 2018.
- Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Scao, T. L., Gugger, S., Drame, M., Lhoest, Q., and Rush, A. M. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP'20)*, pp. 38–45, 2020.
- Xia, W., Qin, C., and Hazan, E. Chain of lora: Efficient fine-tuning of language models via residual learning. *arXiv:2401.04151*, 2024.
- Xu, C., Sun, Q., Zheng, K., Geng, X., Zhao, P., Feng, J., Tao, C., Lin, Q., and Jiang, D. Wizardlm: Empowering large pre-trained language models to follow complex instructions. In *Proceedings of the 12th International Conference on Learning Representations (ICLR'24)*, 2024.
- Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., Zheng, C., Liu, D., and et al. Qwen3 technical report, 2025.
- Yang, Y., Li, X., Zhou, Z., Song, S. L., Wu, J., Nie, L., and Ghanem, B. CorDA: Context-oriented decomposition adaptation of large language models for task-aware parameter-efficient fine-tuning. In *Advances in Neural Information Processing Systems 37 (NeurIPS'24)*, 2024.
- Yaras, C., Wang, P., Balzano, L., and Qu, Q. Compressible dynamics in deep overparameterized low-rank learning & adaptation. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2024.

- Yu, L., Jiang, W., Shi, H., Yu, J., Liu, Z., Zhang, Y., Kwok, J. T., Li, Z., Weller, A., and Liu, W. Metamath: Bootstrap your own mathematical questions for large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR'24)*, 2024.
- Yu, X., Wang, Y., Chen, J., and Xue, L. Altlora: Towards better gradient approximation in low-rank adaptation with alternating projections. In *Advances in Neural Information Processing Systems 38 (NeurIPS'25)*, 2025.
- Zhang, F. and Pilanci, M. Riemannian preconditioned lora for fine-tuning foundation models. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2024.
- Zhang, Q., Chen, M., Bukharin, A., He, P., Cheng, Y., Chen, W., and Zhao, T. Adaptive budget allocation for parameter-efficient fine-tuning. In *Proceedings of the 11th International Conference on Learning Representations (ICLR'23)*, 2023.
- Zhang, Y., Liu, F., and Chen, Y. LoRA-one: One-step full gradient could suffice for fine-tuning large language models, provably and efficiently. In *Proceedings of the 42th International Conference on Machine Learning (ICML'25)*, 2025.
- Zhao, J., Zhang, Z., Chen, B., Wang, Z., Anandkumar, A., and Tian, Y. Galore: Memory-efficient LLM training by gradient low-rank projection. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2024.
- Zheng, L., Chiang, W., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., and Stoica, I. Judging llm-as-a-judge with mt-bench and chatbot arena. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S. (eds.), *Advances in Neural Information Processing Systems 36 (NeurIPS'23)*, 2024a.
- Zheng, T., Zhang, G., Shen, T., Liu, X., Lin, B. Y., Fu, J., Chen, W., and Yue, X. Opencodeinterpreter: Integrating code generation with execution and refinement. In *Findings of the Association for Computational Linguistics (Findings of ACL'24)*, pp. 12834–12859, 2024b.
- Zhu, J., Greenewald, K. H., Nadjahi, K., de Ocariz Borde, H. S., Gabrielsson, R. B., Choshen, L., Ghassemi, M., Yurochkin, M., and Solomon, J. Asymmetry in low-rank adapters of foundation models. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2024.

A. Theoretical Results

A.1. Derivation of the Update Dynamics

By applying the L -smooth inequality in Assumption 3.1, the following result holds:

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) + \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t \rangle + \frac{L}{2} \|\Delta \mathbf{W}_t\|^2. \quad (22)$$

Denote $\Psi(\Delta \mathbf{W}_t) = \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t \rangle + \frac{L}{2} \|\Delta \mathbf{W}_t\|^2$. Take the gradient of $\Psi(\Delta \mathbf{W}_t)$ with respect to $\Delta \mathbf{W}_t$ to find the minimizer:

$$\frac{\partial \Psi(\Delta \mathbf{W}_t)}{\partial \Delta \mathbf{W}_t} = \nabla f(\mathbf{W}_t) + L \Delta \mathbf{W}_t. \quad (23)$$

Setting the gradient to zero yields the optimal solution:

$$\Delta \mathbf{W}_t^* = -\frac{1}{L} \nabla f(\mathbf{W}_t), \quad (24)$$

Substituting $\Delta \mathbf{W}_t^*$ into Eq. (22) gives the following result:

$$\begin{aligned} f(\mathbf{W}_{t+1}) &\leq f(\mathbf{W}_t) + \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t^* \rangle + \frac{L}{2} \|\Delta \mathbf{W}_t^*\|^2 \\ &= f(\mathbf{W}_t) - \frac{1}{2L} \|\nabla f(\mathbf{W}_t)\|^2. \end{aligned} \quad (25)$$

Given Assumption 3.1, we can derive an update rule for $\mathbf{B}_t$ that minimizes a quadratic upper bound of the loss. For the weight matrix $\mathbf{W}_t$, the update rule is $\mathbf{W}_{t+1} = \mathbf{W}_t + \Delta \mathbf{W}_t$, where $\Delta \mathbf{W}_t = \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$. By applying the L -smooth inequality, the following result holds:

$$\begin{aligned} f(\mathbf{W}_{t+1}) &\leq f(\mathbf{W}_t) + \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t \rangle + \frac{L}{2} \|\Delta \mathbf{W}_t\|^2 \\ &= f(\mathbf{W}_t) + \left\langle \nabla f(\mathbf{W}_t), \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\rangle + \frac{L}{2} \left\| \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\|^2. \end{aligned} \quad (26)$$

The objective is to minimize this upper bound in the right-hand side with respect to $\mathbf{B}_t$. Denote

$$\Psi(\mathbf{B}_t) = \left\langle \nabla f(\mathbf{W}_t), \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\rangle + \frac{L}{2} \left\| \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\|^2. \quad (27)$$

The Frobenius norm term can be rewritten according to the trace property:

$$\left\| \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\|^2 = \frac{\alpha^2}{r^2} \text{Tr}(\mathbf{B}_t \mathbf{A}_t \mathbf{A}_t^\top \mathbf{B}_t^\top), \quad (28)$$

where $\top$ denotes matrix transpose. The inner product term can be rewritten as:

$$\begin{aligned} \left\langle \nabla f(\mathbf{W}_t), \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t \right\rangle &= \frac{\alpha}{r} \text{Tr}(\nabla f(\mathbf{W}_t)^\top \mathbf{B}_t \mathbf{A}_t) \\ &= \frac{\alpha}{r} \text{Tr}(\mathbf{A}_t \nabla f(\mathbf{W}_t)^\top \mathbf{B}_t). \end{aligned} \quad (29)$$

Thus, $\Psi(\mathbf{B}_t)$ can be written as:

$$\Psi(\mathbf{B}_t) = \frac{\alpha}{r} \text{Tr}(\mathbf{A}_t \nabla f(\mathbf{W}_t)^\top \mathbf{B}_t) + \frac{L\alpha^2}{2r^2} \text{Tr}(\mathbf{B}_t \mathbf{A}_t \mathbf{A}_t^\top \mathbf{B}_t^\top). \quad (30)$$

Take the gradient of $\Psi(\mathbf{B}_t)$ with respect in order to $\mathbf{B}_t$ to find the minimizer:

$$\frac{\partial \Psi(\mathbf{B}_t)}{\partial \mathbf{B}_t} = \frac{\alpha}{r} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top + \frac{L\alpha^2}{r^2} \mathbf{B}_t \mathbf{A}_t \mathbf{A}_t^\top. \quad (31)$$

Setting the gradient to zero yields the condition for the stationary point:

$$\frac{\alpha}{r} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top + \frac{L\alpha^2}{r^2} \mathbf{B}_t \mathbf{A}_t \mathbf{A}_t^\top = 0. \quad (32)$$

The optimal $\mathbf{B}_t^*$ that minimizes the quadratic upper bound $\Psi(\mathbf{B}_t)$ can be derived as follows:

$$\mathbf{B}_t^* = -\frac{r}{L\alpha} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger, \quad (33)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. This update can be seen as a form of preconditioned gradient step for $\mathbf{B}_t$, where the preconditioner is $(\mathbf{A}_t \mathbf{A}_t^\top)^\dagger$. Let $\mathbf{P}_{\mathbf{A}_t} = \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t$, which is symmetric and idempotent. The updating rule with respect to fixed $\mathbf{A}_t$ is:

$$\begin{aligned} \mathbf{W}_{t+1} &= \mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t \\ &= \mathbf{W}_t - \frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}. \end{aligned} \quad (34)$$

Similarly, the optimal $\mathbf{A}_t^*$ with respect to fixed $\mathbf{B}_t$ is:

$$\mathbf{A}_t^* = -\frac{r}{L\alpha} (\mathbf{B}_t^\top \mathbf{B}_t)^\dagger \mathbf{B}_t^\top \nabla f(\mathbf{W}_t), \quad (35)$$

and the updating rule with respect to fixed $\mathbf{B}_t$ is:

$$\begin{aligned} \mathbf{W}_{t+1} &= \mathbf{W}_t - \frac{1}{L} \mathbf{B}_t (\mathbf{B}_t^\top \mathbf{B}_t)^\dagger \mathbf{B}_t^\top \nabla f(\mathbf{W}_t) \\ &= \mathbf{W}_t - \frac{1}{L} \mathbf{P}_{\mathbf{B}_t} \nabla f(\mathbf{W}_t), \end{aligned} \quad (36)$$

where $\mathbf{P}_{\mathbf{B}_t} = \mathbf{B}_t (\mathbf{B}_t^\top \mathbf{B}_t)^\dagger \mathbf{B}_t^\top$. Substituting $\mathbf{B}_t^*$ into $\Delta \mathbf{W}_t = \frac{\alpha}{r} \mathbf{B}_t \mathbf{A}_t$ gives the following result:

$$\begin{aligned} \Delta \mathbf{W}_t &= -\frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^\dagger \mathbf{A}_t \\ &= -\frac{1}{L} \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}. \end{aligned} \quad (37)$$

Notice that the inner product term in Eq. (26) satisfies:

$$\begin{aligned} \langle \nabla f(\mathbf{W}_t), \Delta \mathbf{W}_t \rangle &= -\frac{1}{L} \langle \nabla f(\mathbf{W}_t), \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \rangle \\ &= -\frac{1}{L} \text{Tr}(\nabla f(\mathbf{W}_t)^\top \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}). \end{aligned} \quad (38)$$

The squared norm term in Eq. (26) satisfies:

$$\begin{aligned} \frac{L}{2} \|\Delta \mathbf{W}_t\|^2 &= \frac{1}{2L} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2 \\ &= \frac{1}{2L} \text{Tr} \left(\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \mathbf{P}_{\mathbf{A}_t}^\top \nabla f(\mathbf{W}_t)^\top \right) \\ &= \frac{1}{2L} \text{Tr} \left(\nabla f(\mathbf{W}_t)^\top \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \right). \end{aligned} \quad (39)$$

Substituting Eq. (38) and Eq. (39) into Eq. (26) yields:

$$\begin{aligned} f(\mathbf{W}_{t+1}) &\leq f(\mathbf{W}_t) - \frac{1}{L} \text{Tr} \left(\nabla f(\mathbf{W}_t)^\top \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \right) \\ &\quad + \frac{1}{2L} \text{Tr} \left(\nabla f(\mathbf{W}_t)^\top \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \right) \\ &= f(\mathbf{W}_t) - \frac{1}{2L} \text{Tr} \left(\nabla f(\mathbf{W}_t)^\top \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \right). \end{aligned} \quad (40)$$

According to Eq. (39), we have:

$$\text{Tr} \left(\nabla f(\mathbf{W}_t)^\top \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \right) = \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2. \quad (41)$$

Therefore, the following result holds:

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) - \frac{1}{2L} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2. \quad (42)$$

A.2. Proof of Theorem 3.3

A key property of ETFs is that they achieve the minimum possible coherence among all unit-norm frame systems, which attains the theoretical lower limit known as the Welch Bound (Welch, 1974; Strohmer & Heath, 2003):

Lemma A.1 (Welch Bound and ETF). *The coherence $\mu(\mathbf{X}) = \max_{i,j} |\mathbf{X}_{:,i}^\top \mathbf{X}_{:,j}|$ of any matrix $\mathbf{X} \in \mathbb{R}^{r \times n}$ with unit-norm columns satisfies the Welch bound:*

$$\mu(\mathbf{X}) \geq \sqrt{\frac{n-r}{r(n-1)}}, \quad (43)$$

where the equality holds if and only if $\mathbf{X}$ is an ETF.

Theorem A.2. *For a given matrix $\mathbf{A}_t \in \mathbb{R}^{r \times n}$, denote the normalized matrix as $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger/2} \mathbf{A}_t$, then the projection matrix can be written as $\mathbf{P}_{\mathbf{A}_t} = \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger} \mathbf{A}_t = \bar{\mathbf{A}}_t^\top \bar{\mathbf{A}}_t$. For gradient $\nabla f(\mathbf{W})$ that has s -effective freedom with $s < 1 + 1/\delta$, denote $\Omega(\bar{\mathbf{A}}_t) = \inf_{\mathbf{W}} \frac{\|\nabla f(\mathbf{W}) \mathbf{P}_{\mathbf{A}_t}\|^2}{\|\nabla f(\mathbf{W})\|^2}$ for $\|\nabla f(\mathbf{W})\| \neq 0$, then the following result holds:*

$$\Omega(\bar{\mathbf{A}}_t) \leq \frac{r}{n} (1 - \epsilon), \quad (44)$$

where $\epsilon = (s-1)\delta$ and $\delta = \sqrt{\frac{n-r}{r(n-1)}}$. The equality holds if and only if $\bar{\mathbf{A}}_t$ is a normalized ETF. In particular, if $\mathbf{A}_t$ is an ETF, then $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger/2} \mathbf{A}_t$ is a normalized ETF.

Proof. Consider $\mathbf{Z} \in \mathbb{R}^{m \times n}$ that satisfy the s -effective freedom property. The energy of the projected signal is:

$$\|\mathbf{Z} \mathbf{P}_{\mathbf{A}_t}\|^2 = \text{tr}(\mathbf{Z} \mathbf{P}_{\mathbf{A}_t} \mathbf{P}_{\mathbf{A}_t}^\top \mathbf{Z}^\top) = \text{tr}(\mathbf{Z} \mathbf{P}_{\mathbf{A}_t} \mathbf{Z}^\top), \quad (45)$$

where the above equality follows from the idempotence and symmetry properties of the projection matrix $\mathbf{P}_{\mathbf{A}_t}$. Denote the normalized matrix as $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{\dagger/2} \mathbf{A}_t \in \mathbb{R}^{r \times n}$, then $\mathbf{P}_{\mathbf{A}_t} = \bar{\mathbf{A}}_t^\top \bar{\mathbf{A}}_t$. Denote the Gram matrix as $\mathbf{G} = \bar{\mathbf{A}}_t^\top \bar{\mathbf{A}}_t$. Let $G_{ii} = \|\bar{\mathbf{A}}_{t:,i}\|_2^2$ be the squared norms of the columns of $\bar{\mathbf{A}}_t$, and define $\kappa(\bar{\mathbf{A}}_t) = \min_{1 \leq i \leq n} G_{ii}$. Let $\mathbf{z}_k$ be the k -th row of $\mathbf{Z}$. Then the following result holds:

$$\begin{aligned} \text{tr}(\mathbf{Z} \mathbf{G} \mathbf{Z}^\top) &= \sum_{k=1}^m \mathbf{z}_k \mathbf{G} \mathbf{z}_k^\top \\ &= \sum_{k=1}^m \left(\sum_{i \in S_k} G_{ii} z_{ki}^2 + \sum_{\substack{i,j \in S_k \\ i \neq j}} G_{ij} z_{ki} z_{kj} \right) \\ &\geq \kappa(\bar{\mathbf{A}}_t) \|\mathbf{Z}\|^2 + \sum_{k=1}^m \sum_{\substack{i,j \in S_k \\ i \neq j}} G_{ij} z_{ki} z_{kj}. \end{aligned} \quad (46)$$

Let $\mu(\bar{\mathbf{A}}_t) = \max_{i \neq j} |\bar{\mathbf{A}}_{t:,i}^\top \bar{\mathbf{A}}_{t:,j}|$ be the coherence of $\bar{\mathbf{A}}_t$. Then $|G_{ij}| \leq \mu(\bar{\mathbf{A}}_t), \forall i \neq j$. Consequently,

$$\left| \sum_{k=1}^m \sum_{\substack{i,j \in S_k \\ i \neq j}} G_{ij} z_{ki} z_{kj} \right| \leq \mu(\bar{\mathbf{A}}_t) \sum_{k=1}^m \sum_{\substack{i,j \in S_k \\ i \neq j}} |z_{ki}| |z_{kj}|. \quad (47)$$

Denote the sum of absolute products over all rows by:

$$S = \sum_{k=1}^m \sum_{\substack{i,j \in S_k \\ i \neq j}} |z_{ki}| |z_{kj}|. \quad (48)$$

For each row k , the following result holds:

$$\begin{aligned} \left(\sum_{i \in S_k} |z_{ki}| \right)^2 &= \sum_{i \in S_k} |z_{ki}|^2 + \sum_{\substack{i,j \in S_k \\ i \neq j}} |z_{ki}| |z_{kj}| \\ &= \|\mathbf{z}_k\|_2^2 + \sum_{\substack{i,j \in S_k \\ i \neq j}} |z_{ki}| |z_{kj}|, \end{aligned} \quad (49)$$

thus, the following result holds:

$$\sum_{\substack{i,j \in S_k \\ i \neq j}} |z_{ki}| |z_{kj}| = \left(\sum_{i \in S_k} |z_{ki}| \right)^2 - \|\mathbf{z}_k\|_2^2. \quad (50)$$

Summing over all rows gives the following result:

$$S = \sum_{k=1}^m \left(\sum_{i \in S_k} |z_{ki}| \right)^2 - \|\mathbf{Z}\|^2. \quad (51)$$

According to the s -effective freedom property, for each k , the following result holds:

$$\left(\sum_{i \in S_k} |z_{ki}| \right)^2 \leq s \| \mathbf{z}_k \|_2^2. \quad (52)$$

Summing over k yields the following result:

$$\sum_{k=1}^m \left(\sum_{i \in S_k} |z_{ki}| \right)^2 \leq s \sum_{k=1}^m \| \mathbf{z}_k \|_2^2 = s \| \mathbf{Z} \|^2. \quad (53)$$

Substituting this into the expression for S gives:

$$S \leq s \| \mathbf{Z} \|^2 - \| \mathbf{Z} \|^2 = (s-1) \| \mathbf{Z} \|^2. \quad (54)$$

This directly implies the following result:

$$\left| \sum_{k=1}^m \sum_{\substack{i,j \in S_k \\ i \neq j}} G_{ij} z_{ki} z_{kj} \right| \leq \mu(\bar{\mathbf{A}}_t) (s-1) \| \mathbf{Z} \|^2. \quad (55)$$

Thus, the following result holds:

$$\text{tr}(\mathbf{Z} \mathbf{G} \mathbf{Z}^\top) \geq \kappa(\bar{\mathbf{A}}_t) \| \mathbf{Z} \|^2 - \mu(\bar{\mathbf{A}}_t) (s-1) \| \mathbf{Z} \|^2, \quad (56)$$

which yields the lower bound:

$$\| \mathbf{Z} \mathbf{P}_{\mathbf{A}_t} \|^2 \geq (\kappa(\bar{\mathbf{A}}_t) - (s-1) \mu(\bar{\mathbf{A}}_t)) \| \mathbf{Z} \|^2. \quad (57)$$

Let $\Omega(\bar{\mathbf{A}}_t) = \inf \frac{\| \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \|^2}{\| \nabla f(\mathbf{W}_t) \|^2}$ for $\| \nabla f(\mathbf{W}_t) \| \neq 0$, then $\Omega(\bar{\mathbf{A}}_t) = \kappa(\bar{\mathbf{A}}_t) - (s-1) \mu(\bar{\mathbf{A}}_t)$. Maximizing $\Omega(\bar{\mathbf{A}}_t)$ is equal to maximizing $\kappa(\bar{\mathbf{A}}_t)$ while minimizing $\mu(\bar{\mathbf{A}}_t)$. Notice that $\mathbf{G} = \mathbf{P}_{\mathbf{A}_t}$ is a projection matrix; its singular values are either 0 or 1. Thus, $\text{tr}(\mathbf{G}) = \sum_{i=1}^n G_{ii}$ equals the rank of $\mathbf{G}$, which is at most r , i.e.,

$$\text{tr}(\mathbf{G}) = \sum_{i=1}^n G_{ii} \leq r. \quad (58)$$

To maximize $\kappa(\bar{\mathbf{A}}_t)$ under the constraint $\sum_i G_{ii} \leq r$, all diagonal entries must be equal, i.e., $\kappa(\bar{\mathbf{A}}_t) = G_{ii} = \frac{r}{n}$ for $\forall i$. On the other hand, Lemma A.1 states that the coherence $\mu(\bar{\mathbf{A}}_t)$ is minimized (the Welch bound is attained) if and only if $\bar{\mathbf{A}}_t$ is an ETF (with a scaling factor). Denote $\bar{\mathbf{A}}_t = \alpha \mathbf{X}$, where $\mathbf{X} \in \mathbb{R}^{r \times n}$ is an ETF, and $\alpha > 0$ is a scaling factor to be determined. From $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{-1/2} \mathbf{A}_t$ it follows that $\mathbf{A}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \bar{\mathbf{A}}_t = \alpha (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \mathbf{X}$. Notice that:

$$\begin{aligned} \mathbf{A}_t \mathbf{A}_t^\top &= \left(\alpha (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \mathbf{X} \right) \left(\alpha (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \mathbf{X} \right)^\top \\ &= \alpha^2 (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \mathbf{X} \mathbf{X}^\top (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \\ &= \alpha^2 (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \left(\frac{n}{r} \mathbf{I}_r \right) (\mathbf{A}_t \mathbf{A}_t^\top)^{1/2} \\ &= \alpha^2 \frac{n}{r} (\mathbf{A}_t \mathbf{A}_t^\top). \end{aligned} \quad (59)$$

Thus $\alpha^2 \frac{n}{r} = 1$, i.e., $\alpha = \sqrt{\frac{r}{n}}$. Therefore, $\bar{\mathbf{A}}_t = \sqrt{\frac{r}{n}} \mathbf{X}$ is a normalized ETF, i.e., $\| \bar{\mathbf{A}}_{t:,i} \|_2 = \sqrt{\frac{r}{n}}$, $\forall i \in \{1, \dots, n\}$, $\mu(\bar{\mathbf{A}}_t) = | \bar{\mathbf{A}}_{t:,i}^\top \bar{\mathbf{A}}_{t:,j} | = \frac{r}{n} \sqrt{\frac{n-r}{r(n-1)}}$, $\forall i \neq j$, and $\bar{\mathbf{A}}_t \bar{\mathbf{A}}_t^\top = \mathbf{I}_r$. Note that the condition $\| \bar{\mathbf{A}}_{t:,i} \|_2 = \sqrt{\frac{r}{n}}$ satisfies the requirement for maximizing $\kappa(\bar{\mathbf{A}}_t)$, i.e., $G_{ii} = \frac{r}{n}$. With maximized $\kappa(\bar{\mathbf{A}}_t) = \frac{r}{n}$ and minimized $\mu(\bar{\mathbf{A}}_t) = \frac{r}{n} \sqrt{\frac{n-r}{r(n-1)}}$, we can derive the following result:

$$\max_{\bar{\mathbf{A}}_t} \Omega(\bar{\mathbf{A}}_t) = \frac{r}{n} - (s-1) \cdot \frac{r}{n} \sqrt{\frac{n-r}{r(n-1)}}, \quad (60)$$

which leads to the following result:

$$\max_{\bar{\mathbf{A}}_t} \Omega(\bar{\mathbf{A}}_t) = \frac{r}{n} (1 - \epsilon), \quad (61)$$

where $\epsilon = (s-1) \sqrt{\frac{n-r}{r(n-1)}}$. Let $\hat{\mathbf{A}}_t = \arg \max_{\bar{\mathbf{A}}_t} \Omega(\bar{\mathbf{A}}_t)$, then $\hat{\mathbf{A}}_t$ is a normalized ETF. Thus, by applying $\mathbf{Z}$ with $\nabla f(\mathbf{W}_t)$, Eq. (57) provides the following best possible lower bound:

$$\begin{aligned} &\| \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \|^2 \\ &\geq \left(\frac{r}{n} - (s-1) \cdot \frac{r}{n} \sqrt{\frac{n-r}{r(n-1)}} \right) \| \nabla f(\mathbf{W}_t) \|^2 \\ &= \frac{r}{n} \left(1 - (s-1) \sqrt{\frac{n-r}{r(n-1)}} \right) \| \nabla f(\mathbf{W}_t) \|^2, \end{aligned} \quad (62)$$

which is achieved when $\bar{\mathbf{A}}_t$ is a normalized ETF. Let $\mathbf{A}_t$ be an ETF. By definition, $\| \mathbf{A}_{t:,i} \|_2 = 1$, $\forall i$, $| \mathbf{A}_{t:,i}^\top \mathbf{A}_{t:,j} | = \sqrt{\frac{n-r}{r(n-1)}}$, $\forall i \neq j$, and $\mathbf{A}_t \mathbf{A}_t^\top = \frac{n}{r} \mathbf{I}_r$. Since $\mathbf{A}_t \mathbf{A}_t^\top = \frac{n}{r} \mathbf{I}_r$, we have that $(\mathbf{A}_t \mathbf{A}_t^\top)^{-1/2} = \sqrt{\frac{r}{n}} \mathbf{I}_r$. Therefore, $\bar{\mathbf{A}}_t = (\mathbf{A}_t \mathbf{A}_t^\top)^{-1/2} \mathbf{A}_t = \sqrt{\frac{r}{n}} \mathbf{A}_t$. Now we verify the conditions for $\bar{\mathbf{A}}_t$. (1) $\forall i$, $\| \bar{\mathbf{A}}_{t:,i} \|_2 = \sqrt{\frac{r}{n}} \| \mathbf{A}_{t:,i} \|_2 = \sqrt{\frac{r}{n}}$. (2) $\forall i \neq j$, $| \bar{\mathbf{A}}_{t:,i}^\top \bar{\mathbf{A}}_{t:,j} | = \frac{r}{n} | \mathbf{A}_{t:,i}^\top \mathbf{A}_{t:,j} | = \frac{r}{n} \sqrt{\frac{n-r}{r(n-1)}}$. (3) $\bar{\mathbf{A}}_t \bar{\mathbf{A}}_t^\top = \left(\sqrt{\frac{r}{n}} \mathbf{A}_t \right) \left(\sqrt{\frac{r}{n}} \mathbf{A}_t^\top \right) = \frac{r}{n} \mathbf{A}_t \mathbf{A}_t^\top = \mathbf{I}_r$. Thus $\bar{\mathbf{A}}_t$ fulfills all required properties, completing the proof. $\square$

A.3. Proof of Theorem 3.4

Theorem A.3. Under Assumption 3.1, if the matrix $\mathbf{A}_t$ is an ETF and $\eta_B = \frac{r}{L\alpha}$ in Algorithm 1, then the following result holds for Algorithm 1:

$$\frac{1}{T} \sum_{t=0}^{T-1} \| \nabla f(\mathbf{W}_t) \|^2 \leq \frac{2nL}{r(1-\epsilon)T} (f(\mathbf{W}_0) - f^*), \quad (63)$$

where $\epsilon = (s-1) \sqrt{\frac{n-r}{r(n-1)}}$, and $f^* = \inf_{\mathbf{W}} f(\mathbf{W})$.

Proof. Denote $\Delta = f(\mathbf{W}_0) - f^*$. According to the descent inequality in Eq. (42), for each step t :

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) - \frac{1}{2L} \| \nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t} \|^2. \quad (64)$$

Summing over $t = 0, 1, \dots, T-1$ yields:

$$\frac{1}{2L} \sum_{t=0}^{T-1} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2 \leq f(\mathbf{W}_0) - f(\mathbf{W}_T) \leq \Delta. \quad (65)$$

Therefore, the following result holds:

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2 \leq \frac{2L\Delta}{T}, \quad (66)$$

which leads to the following result:

$$\min_{0 \leq t < T} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2 \leq \frac{2L\Delta}{T}. \quad (67)$$

This gives a convergence rate in terms of the squared norm of the projected gradient. Note that Theorem 3.3 states that if the gradient matrix has s -effective freedom with $s < 1 + \sqrt{\frac{r(n-1)}{n-r}}$ and $\mathbf{A}_t$ is an ETF, then

$$\|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2 \geq \frac{r}{n} (1 - \epsilon) \|\nabla f(\mathbf{W}_t)\|^2, \quad (68)$$

where $\epsilon = (s-1)\sqrt{\frac{n-r}{r(n-1)}}$. Then:

$$\|\nabla f(\mathbf{W}_t)\|^2 \leq \frac{n}{r(1-\epsilon)} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2, \quad (69)$$

and consequently, the following result holds:

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla f(\mathbf{W}_t)\|^2 \leq \frac{2nL\Delta}{r(1-\epsilon)T}. \quad (70)$$

□

A.4. Proof of Theorem 3.6

Theorem A.4. Under Assumption 3.1 and 3.5, if the matrix $\mathbf{A}_t$ is an ETF and $\eta_{\mathbf{B}} = \frac{r}{L\alpha}$ in Algorithm 1, then the following result holds for Algorithm 1:

$$f(\mathbf{W}_t) - f^* \leq \left(1 - \frac{\mu r}{nL} (1 - \epsilon)\right)^t (f(\mathbf{W}_0) - f^*), \quad (71)$$

where $\epsilon = (s-1)\sqrt{\frac{n-r}{r(n-1)}}$, and $f^* = \inf_{\mathbf{W}} f(\mathbf{W})$.

Proof. According to the descent inequality in Eq. (42), for each step t , the following result holds:

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) - \frac{1}{2L} \|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2. \quad (72)$$

Note that Theorem 3.3 states that if the gradient matrix has s -effective freedom with $s < 1 + \sqrt{\frac{r(n-1)}{n-r}}$ and $\mathbf{A}_t$ is an ETF, then the following result holds:

$$\|\nabla f(\mathbf{W}_t) \mathbf{P}_{\mathbf{A}_t}\|^2 \geq \frac{r}{n} (1 - \epsilon) \|\nabla f(\mathbf{W}_t)\|^2, \quad (73)$$

where $\epsilon = (s-1)\sqrt{\frac{n-r}{r(n-1)}}$. Substituting the lower bound on the projected energy yields:

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) - \frac{r}{2nL} (1 - \epsilon) \|\nabla f(\mathbf{W}_t)\|^2. \quad (74)$$

By applying the PL condition formulation in Assumption 3.5, i.e., $\frac{1}{2} \|\nabla f(\mathbf{W}_t)\|^2 \geq \mu(f(\mathbf{W}_t) - f^*)$, the following result holds:

$$f(\mathbf{W}_{t+1}) \leq f(\mathbf{W}_t) - \frac{\mu r}{nL} (1 - \epsilon) (f(\mathbf{W}_t) - f^*). \quad (75)$$

Rearranging yields the following result:

$$f(\mathbf{W}_{t+1}) - f^* \leq f(\mathbf{W}_t) - f^* - \frac{\mu r}{nL} (1 - \epsilon) (f(\mathbf{W}_t) - f^*), \quad (76)$$

which leads to the following result:

$$f(\mathbf{W}_{t+1}) - f^* \leq \left(1 - \frac{\mu r}{nL} (1 - \epsilon)\right) (f(\mathbf{W}_t) - f^*). \quad (77)$$

Therefore, the following result holds:

$$f(\mathbf{W}_t) - f^* \leq \left(1 - \frac{\mu r}{nL} (1 - \epsilon)\right)^t (f(\mathbf{W}_0) - f^*). \quad (78)$$

□

B. Experimental Results

B.1. Construction Approach of ETF

In practice, the LoRA matrix $\mathbf{A} \in \mathbb{R}^{r \times n}$ satisfies $r \ll n$, therefore we provide a numerical optimization approach for constructing the ETFs on a large scale as follows. We first generate an optimal Grassmannian frame $\mathbf{X} \in \mathbb{R}^{r \times n}$ by maximizing the separation between the lines spanned by its column vectors in an iterative way following (Tahir et al., 2019; Meenakshi & Srirangarajan, 2021; Isaacs, 2011). This iterative method minimizes a loss function that penalizes deviations between the off-diagonal magnitudes of the matrix and the Welch bound (Welch, 1974). Specifically, let $\mathbf{G} = \mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{n \times n}$ denote the Gram matrix, the loss function takes the form $\mathcal{L}(\mathbf{X}) = \sum_{i \neq j} (|G_{ij}| - \delta)^2$, where $\delta = \sqrt{\frac{n-r}{r(n-1)}}$ is the Welch bound. Additionally, we enforce the following constraints: $\|\mathbf{X}_{:,j}\|_2 = 1, \forall j \in \{1, \dots, n\}$ and $\mathbf{X}\mathbf{X}^\top = \frac{n}{r}\mathbf{I}_r$. We iteratively update $\mathbf{X}$ to minimize $\mathcal{L}(\mathbf{X})$ while projecting onto the constraint set until the off-diagonal magnitudes are sufficiently close to δ . Upon reaching the bound, the algorithm successfully yields an approximate ETF as that in (Fickus & Mixon, 2015; Strohmer & Heath, 2003). According to the property of unitary equivalence, for any random orthogonal matrix $\mathbf{U} \in \mathbb{R}^{r \times r}$, the transformed matrix $\mathbf{A} = \mathbf{U}\mathbf{X}$ is also an approximate ETF. Therefore, we can generate distinct ETFs by multiplying the base frame $\mathbf{X}$ with random orthogonal matrices $\mathbf{U}$.

B.2. Natural Language Understanding Tasks

B.2.1. SETUP

For natural language understanding tasks, we evaluate these methods on several GLUE benchmark datasets (Wang et al., 2019), *i.e.*, CoLA (Warstadt et al., 2019), SST-2 (Socher et al., 2013), MRPC (Dolan & Brockett, 2005), QQP², MNLI (Williams et al., 2018), QNLI (Rajpurkar et al., 2016), and RTE (Wang et al., 2019).

We use the RoBERTa (Liu et al., 2019) model and the Llama-3.1-8B (Dubey et al., 2024) model implemented by the HuggingFace Transformers library (Wolf et al., 2020) as the backbone. For Llama-3.1-8B, we use the FP16/BF16 mixed-precision strategy to reduce the GPU memory consumption. All the experiments are conducted on a platform with 8 NVIDIA GeForce RTX 4090 GPUs.

For the RoBERTa model, we adopt the AdamW optimizer (Loshchilov & Hutter, 2019) and report the best result with the learning rate $\eta \in \{5 \times 10^{-4}, 4 \times 10^{-4}, 3 \times 10^{-4}, 2 \times 10^{-4}, 1 \times 10^{-4}\}$ with cosine annealing for the learning rate schedule. We apply the LoRA adapters to multi-head attention layers and feed-forward layers following (Zhao

et al., 2024). The batch size is 32, the number of epochs is 30, the gradient accumulation step is 1, the scaling factor is twice the rank, and the maximum sequence length is 128.

For the Llama-3.1-8B model, we adopt the AdamW optimizer (Loshchilov & Hutter, 2019) and report the best result with the learning rate $\eta \in \{5 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-4}, 5 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-5}\}$ with cosine annealing for the learning rate schedule. We apply the LoRA adapters to multi-head attention layers following (Hu et al., 2022). The batch size is 4, the number of epochs is 3, the gradient accumulation step is 1, the scaling factor is twice the rank, and the maximum sequence length is 128.

B.2.2. ADDITIONAL RESULTS

We repeat all experiments 5 times and report the average accuracy along with the standard deviation. For the RoBERTa model, results are presented for LoRA ranks 4, 8, and 16 in Table 4, Table 1, and Table 5, respectively. Similarly, for the Llama-3.1-8B model, results for LoRA ranks 4, 8, and 16 are reported in Table 6, Table 7, and Table 8, respectively. The results show that the LoRA variants with ETF consistently outperform their default initialization counterparts across all datasets. This result validates our theoretical result in Theorem 3.3, further showing that ETF can effectively facilitate effective gradient descent and enhance the fine-tuning performance.

Comparisons of Galore with and without ETF of RoBERTa and Llama-3.1-8B for rank 4, 8, and 16 are shown in Table 2 and Table 9, respectively. The results show that ETF-initialized GaLore achieves comparable performance to its default SVD initialization.

We depict the convergence curves for these methods of the RoBERTa model for rank 4, 8, 16 in Figure 3, 4, 5. We omit the convergence curves of LoRA-GA and PiSSA, as they modify the initialization approach of LoRA. Therefore, their ETF-initialization versions exhibit the same performance as the ETF-initialization version of LoRA. The results show that these methods with ETF initialization converge faster than their default initialization counterparts.

²data.quora.com/First-Quora-Dataset-Release-Question-Pairs

Table 4. Test accuracy(%) of RoBERTa with LoRA variants (with/without ETF) on GLUE benchmark datasets (rank 4).

Method	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
LoRA	w/o	57.88 $\pm$ 1.25	93.77 $\pm$ 0.14	86.91 $\pm$ 0.83	89.37 $\pm$ 0.03	85.97 $\pm$ 0.05	91.53 $\pm$ 0.23	67.75 $\pm$ 1.19
	w/	59.15 $\pm$ 0.57 (+1.27)	94.95 $\pm$ 0.05 (+1.18)	87.99 $\pm$ 0.31 (+1.08)	90.53 $\pm$ 0.02 (+1.16)	87.02 $\pm$ 0.03 (+1.05)	92.58 $\pm$ 0.07 (+1.05)	71.36 $\pm$ 0.95 (+3.61)
LoRA-GA	w/o	57.45 $\pm$ 1.40	94.04 $\pm$ 0.25	86.77 $\pm$ 0.60	90.38 $\pm$ 0.05	86.55 $\pm$ 0.05	92.38 $\pm$ 0.14	70.40 $\pm$ 0.50
	w/	59.15 $\pm$ 0.57 (+1.70)	94.95 $\pm$ 0.05 (+0.91)	87.99 $\pm$ 0.31 (+1.22)	90.53 $\pm$ 0.02 (+0.15)	87.02 $\pm$ 0.03 (+0.47)	92.58 $\pm$ 0.07 (+0.20)	71.36 $\pm$ 0.95 (+0.96)
PiSSA	w/o	58.46 $\pm$ 0.52	94.57 $\pm$ 0.11	87.50 $\pm$ 0.35	90.42 $\pm$ 0.42	86.24 $\pm$ 0.09	92.26 $\pm$ 0.26	71.15 $\pm$ 1.53
	w/	59.15 $\pm$ 0.57 (+0.69)	94.95 $\pm$ 0.05 (+0.38)	87.99 $\pm$ 0.31 (+0.49)	90.53 $\pm$ 0.02 (+0.11)	87.02 $\pm$ 0.03 (+0.78)	92.58 $\pm$ 0.07 (+0.32)	71.36 $\pm$ 0.95 (+0.21)
FLORA	w/o	58.47 $\pm$ 1.06	93.62 $\pm$ 0.19	87.49 $\pm$ 0.70	90.22 $\pm$ 0.03	85.75 $\pm$ 0.02	91.71 $\pm$ 0.08	69.92 $\pm$ 0.74
	w/	60.90 $\pm$ 0.27 (+2.43)	94.99 $\pm$ 0.14 (+1.37)	88.52 $\pm$ 0.39 (+1.03)	91.30 $\pm$ 0.01 (+1.08)	87.06 $\pm$ 0.19 (+1.31)	92.80 $\pm$ 0.04 (+1.09)	74.01 $\pm$ 1.79 (+4.09)
AsymLoRA	w/o	58.55 $\pm$ 0.36	93.91 $\pm$ 0.11	86.11 $\pm$ 0.23	89.03 $\pm$ 0.05	85.78 $\pm$ 0.10	91.45 $\pm$ 0.17	69.19 $\pm$ 0.61
	w/	58.86 $\pm$ 0.22 (+0.31)	94.92 $\pm$ 0.24 (+1.01)	87.66 $\pm$ 0.58 (+1.55)	90.27 $\pm$ 0.03 (+1.24)	86.88 $\pm$ 0.02 (+1.10)	92.53 $\pm$ 0.08 (+1.08)	73.41 $\pm$ 1.33 (+4.22)
ReLoRA	w/o	54.18 $\pm$ 0.12	92.09 $\pm$ 0.19	86.27 $\pm$ 0.35	88.29 $\pm$ 0.04	81.76 $\pm$ 0.15	89.88 $\pm$ 0.27	68.71 $\pm$ 0.45
	w/	57.37 $\pm$ 1.06 (+3.19)	93.92 $\pm$ 0.23 (+1.83)	87.83 $\pm$ 0.50 (+1.56)	90.17 $\pm$ 0.04 (+1.88)	84.31 $\pm$ 0.05 (+2.55)	91.17 $\pm$ 0.10 (+1.29)	70.40 $\pm$ 0.36 (+1.69)
RAC-LoRA	w/o	56.60 $\pm$ 1.51	93.70 $\pm$ 0.14	86.52 $\pm$ 0.35	89.51 $\pm$ 0.06	85.83 $\pm$ 0.04	91.22 $\pm$ 0.07	72.80 $\pm$ 1.90
	w/	59.09 $\pm$ 0.21 (+2.49)	94.72 $\pm$ 0.09 (+1.02)	88.15 $\pm$ 0.31 (+1.63)	90.52 $\pm$ 0.02 (+1.01)	86.99 $\pm$ 0.02 (+1.16)	92.42 $\pm$ 0.13 (+1.20)	73.89 $\pm$ 2.07 (+1.09)

Table 5. Test accuracy(%) of RoBERTa with LoRA variants (with/without ETF) on GLUE benchmark datasets (rank 16).

Method	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
LoRA	w/o	57.64 $\pm$ 0.46	94.16 $\pm$ 0.05	87.17 $\pm$ 0.61	90.34 $\pm$ 0.01	86.36 $\pm$ 0.06	91.78 $\pm$ 0.05	71.84 $\pm$ 0.29
	w/	59.43 $\pm$ 0.61 (+1.79)	95.18 $\pm$ 0.27 (+1.02)	88.89 $\pm$ 0.31 (+1.72)	91.37 $\pm$ 0.07 (+1.03)	87.38 $\pm$ 0.07 (+1.02)	92.89 $\pm$ 0.04 (+1.11)	74.73 $\pm$ 1.18 (+2.89)
LoRA-GA	w/o	57.98 $\pm$ 0.11	94.04 $\pm$ 0.39	87.42 $\pm$ 0.31	91.19 $\pm$ 0.30	87.13 $\pm$ 0.09	92.27 $\pm$ 0.08	72.80 $\pm$ 0.61
	w/	59.43 $\pm$ 0.61 (+1.45)	95.18 $\pm$ 0.27 (+1.14)	88.89 $\pm$ 0.31 (+1.47)	91.37 $\pm$ 0.07 (+0.18)	87.38 $\pm$ 0.07 (+0.25)	92.89 $\pm$ 0.04 (+0.62)	74.73 $\pm$ 1.18 (+1.93)
PiSSA	w/o	59.23 $\pm$ 0.99	94.30 $\pm$ 0.14	88.64 $\pm$ 0.95	91.31 $\pm$ 0.03	85.94 $\pm$ 0.04	92.07 $\pm$ 0.04	74.37 $\pm$ 1.84
	w/	59.43 $\pm$ 0.61 (+0.20)	95.18 $\pm$ 0.27 (+0.88)	88.89 $\pm$ 0.31 (+0.25)	91.37 $\pm$ 0.07 (+0.06)	87.38 $\pm$ 0.07 (+1.44)	92.89 $\pm$ 0.04 (+0.82)	74.73 $\pm$ 1.18 (+0.36)
FLORA	w/o	61.97 $\pm$ 0.83	94.27 $\pm$ 0.52	87.08 $\pm$ 0.05	90.51 $\pm$ 0.03	86.64 $\pm$ 0.25	91.93 $\pm$ 0.07	75.45 $\pm$ 0.81
	w/	63.90 $\pm$ 1.52 (+1.93)	95.49 $\pm$ 0.24 (+1.22)	88.64 $\pm$ 0.11 (+1.56)	91.59 $\pm$ 0.03 (+1.08)	87.65 $\pm$ 0.05 (+1.01)	92.95 $\pm$ 0.09 (+1.02)	76.77 $\pm$ 0.85 (+1.32)
AsymLoRA	w/o	61.14 $\pm$ 0.89	94.01 $\pm$ 0.06	86.85 $\pm$ 0.42	90.09 $\pm$ 0.06	85.94 $\pm$ 0.18	91.50 $\pm$ 0.05	73.29 $\pm$ 1.53
	w/	62.49 $\pm$ 0.30 (+1.35)	95.02 $\pm$ 0.05 (+1.01)	88.97 $\pm$ 0.69 (+2.12)	91.30 $\pm$ 0.05 (+1.21)	87.00 $\pm$ 0.13 (+1.06)	92.54 $\pm$ 0.22 (+1.04)	76.17 $\pm$ 1.04 (+2.88)
ReLoRA	w/o	58.86 $\pm$ 0.97	91.97 $\pm$ 0.37	86.68 $\pm$ 0.42	89.91 $\pm$ 0.02	83.08 $\pm$ 0.21	90.53 $\pm$ 0.17	70.88 $\pm$ 1.90
	w/	60.28 $\pm$ 0.63 (+1.42)	93.77 $\pm$ 0.11 (+1.80)	87.88 $\pm$ 0.90 (+1.20)	91.07 $\pm$ 0.09 (+1.16)	85.18 $\pm$ 0.13 (+2.10)	91.54 $\pm$ 0.16 (+1.01)	73.04 $\pm$ 1.39 (+2.16)
RAC-LoRA	w/o	59.89 $\pm$ 0.98	94.00 $\pm$ 0.16	87.91 $\pm$ 0.31	90.13 $\pm$ 0.02	86.05 $\pm$ 0.06	91.58 $\pm$ 0.23	73.89 $\pm$ 1.62
	w/	61.52 $\pm$ 0.95 (+1.63)	95.06 $\pm$ 0.19 (+1.06)	88.97 $\pm$ 0.24 (+1.06)	91.17 $\pm$ 0.09 (+1.04)	87.10 $\pm$ 0.11 (+1.05)	92.60 $\pm$ 0.04 (+1.02)	74.91 $\pm$ 1.27 (+1.02)

Table 6. Test accuracy(%) of Llama-3.1-8B with LoRA variants (with/without ETF) on GLUE benchmark datasets (rank 4).

Method	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
LoRA	w/o	67.77 $\pm$ 0.73	96.62 $\pm$ 0.06	86.15 $\pm$ 1.35	90.30 $\pm$ 0.01	91.36 $\pm$ 0.10	95.22 $\pm$ 0.15	83.39 $\pm$ 0.72
	w/	68.98 $\pm$ 0.46 (+1.21)	96.79 $\pm$ 0.04 (+0.17)	86.27 $\pm$ 1.59 (+0.12)	90.36 $\pm$ 0.03 (+0.06)	91.47 $\pm$ 0.11 (+0.11)	95.26 $\pm$ 0.10 (+0.04)	87.24 $\pm$ 1.23 (+3.85)
LoRA-GA	w/o	67.99 $\pm$ 0.78	96.73 $\pm$ 0.06	87.08 $\pm$ 0.74	90.81 $\pm$ 0.03	91.27 $\pm$ 0.02	95.11 $\pm$ 0.02	86.46 $\pm$ 0.36
	w/	68.98 $\pm$ 0.46 (+0.99)	96.79 $\pm$ 0.04 (+0.06)	86.27 $\pm$ 1.59 (-0.81)	90.36 $\pm$ 0.03 (-0.45)	91.47 $\pm$ 0.11 (+0.20)	95.26 $\pm$ 0.10 (+0.15)	87.24 $\pm$ 1.23 (+0.78)
PiSSA	w/o	67.57 $\pm$ 0.18	96.76 $\pm$ 0.23	86.52 $\pm$ 0.98	90.26 $\pm$ 0.08	91.41 $\pm$ 0.02	95.13 $\pm$ 0.05	82.13 $\pm$ 0.54
	w/	68.98 $\pm$ 0.46 (+1.41)	96.79 $\pm$ 0.04 (+0.03)	86.27 $\pm$ 1.59 (-0.25)	90.36 $\pm$ 0.03 (+0.10)	91.47 $\pm$ 0.11 (+0.06)	95.26 $\pm$ 0.10 (+0.13)	87.24 $\pm$ 1.23 (+5.11)
FLORA	w/o	66.61 $\pm$ 0.21	96.06 $\pm$ 0.08	83.72 $\pm$ 1.23	90.04 $\pm$ 0.36	90.23 $\pm$ 0.04	95.18 $\pm$ 0.10	80.14 $\pm$ 1.38
	w/	68.75 $\pm$ 0.05 (+2.14)	96.79 $\pm$ 0.02 (+0.73)	84.56 $\pm$ 1.24 (+0.84)	90.25 $\pm$ 0.16 (+0.21)	90.40 $\pm$ 0.05 (+0.17)	95.27 $\pm$ 0.06 (+0.09)	83.57 $\pm$ 0.94 (+3.43)
AsymLoRA	w/o	50.95 $\pm$ 1.24	95.24 $\pm$ 0.06	71.08 $\pm$ 0.49	88.19 $\pm$ 0.02	89.56 $\pm$ 0.17	90.79 $\pm$ 0.29	63.18 $\pm$ 0.45
	w/	53.51 $\pm$ 1.88 (+2.56)	95.18 $\pm$ 0.03 (-0.06)	73.16 $\pm$ 1.35 (+2.08)	88.85 $\pm$ 0.01 (+0.66)	90.38 $\pm$ 0.08 (+0.82)	92.55 $\pm$ 0.06 (+1.76)	66.06 $\pm$ 0.36 (+2.88)
ReLoRA	w/o	65.72 $\pm$ 0.32	95.81 $\pm$ 0.06	86.52 $\pm$ 1.96	88.06 $\pm$ 0.07	89.81 $\pm$ 0.02	93.36 $\pm$ 0.19	83.03 $\pm$ 0.36
	w/	65.61 $\pm$ 0.25 (-0.11)	96.22 $\pm$ 0.02 (+0.41)	87.13 $\pm$ 1.10 (+0.61)	89.03 $\pm$ 0.05 (+0.97)	89.85 $\pm$ 0.46 (+0.04)	93.85 $\pm$ 0.05 (+0.49)	83.35 $\pm$ 1.27 (+0.32)
RAC-LoRA	w/o	51.09 $\pm$ 1.42	95.07 $\pm$ 0.23	71.69 $\pm$ 0.86	88.18 $\pm$ 0.03	88.67 $\pm$ 0.09	90.45 $\pm$ 0.25	70.68 $\pm$ 1.26
	w/	56.60 $\pm$ 1.26 (+5.51)	95.64 $\pm$ 0.07 (+0.57)	73.28 $\pm$ 0.74 (+1.59)	88.76 $\pm$ 0.01 (+0.58)	89.85 $\pm$ 0.02 (+1.18)	92.68 $\pm$ 0.28 (+2.23)	71.46 $\pm$ 0.45 (+0.78)

Table 7. Test accuracy(%) of Llama-3.1-8B with LoRA variants (with/without ETF) on GLUE benchmark datasets (rank 8).

Method	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
LoRA	w/o	67.56 $\pm$ 0.23	95.46 $\pm$ 0.05	87.75 $\pm$ 0.35	90.54 $\pm$ 0.11	90.34 $\pm$ 0.03	94.19 $\pm$ 0.05	84.44 $\pm$ 0.85
	w/	68.75 $\pm$ 0.20 (+1.19)	96.44 $\pm$ 0.18 (+0.98)	89.95 $\pm$ 0.44 (+2.20)	91.56 $\pm$ 0.20 (+1.02)	91.57 $\pm$ 0.08 (+1.23)	95.21 $\pm$ 0.05 (+1.02)	86.48 $\pm$ 0.10 (+2.04)
LoRA-GA	w/o	67.27 $\pm$ 0.88	95.22 $\pm$ 0.12	88.89 $\pm$ 0.69	90.49 $\pm$ 0.86	90.64 $\pm$ 0.08	94.42 $\pm$ 0.04	86.20 $\pm$ 1.30
	w/	68.75 $\pm$ 0.20 (+1.48)	96.44 $\pm$ 0.18 (+1.22)	89.95 $\pm$ 0.44 (+1.06)	91.56 $\pm$ 0.20 (+1.07)	91.57 $\pm$ 0.08 (+0.93)	95.21 $\pm$ 0.05 (+0.79)	86.48 $\pm$ 0.10 (+0.28)
PiSSA	w/o	66.86 $\pm$ 1.42	94.90 $\pm$ 0.25	87.50 $\pm$ 0.31	90.45 $\pm$ 0.06	90.47 $\pm$ 0.09	94.07 $\pm$ 0.16	85.92 $\pm$ 1.33
	w/	68.75 $\pm$ 0.20 (+1.89)	96.44 $\pm$ 0.18 (+1.54)	89.95 $\pm$ 0.44 (+2.45)	91.56 $\pm$ 0.20 (+1.11)	91.57 $\pm$ 0.08 (+1.10)	95.21 $\pm$ 0.05 (+1.14)	86.48 $\pm$ 0.10 (+0.56)
FLORA	w/o	65.87 $\pm$ 1.38	95.56 $\pm$ 0.06	87.99 $\pm$ 0.98	89.68 $\pm$ 0.05	89.22 $\pm$ 0.11	93.39 $\pm$ 0.05	85.30 $\pm$ 0.54
	w/	67.30 $\pm$ 1.77 (+1.43)	96.90 $\pm$ 0.03 (+1.34)	89.95 $\pm$ 0.12 (+1.96)	90.99 $\pm$ 0.05 (+1.31)	91.41 $\pm$ 0.23 (+2.19)	94.98 $\pm$ 0.29 (+1.59)	86.32 $\pm$ 1.81 (+1.02)
AsymLoRA	w/o	55.65 $\pm$ 1.10	94.42 $\pm$ 0.05	75.00 $\pm$ 0.81	88.07 $\pm$ 0.01	89.99 $\pm$ 0.03	92.38 $\pm$ 0.08	73.18 $\pm$ 1.87
	w/	60.69 $\pm$ 1.87 (+5.04)	95.54 $\pm$ 0.06 (+1.12)	77.22 $\pm$ 0.12 (+2.22)	89.44 $\pm$ 0.06 (+1.37)	90.94 $\pm$ 0.01 (+0.95)	93.46 $\pm$ 0.12 (+1.08)	75.08 $\pm$ 1.43 (+1.90)
ReLoRA	w/o	66.31 $\pm$ 0.56	95.33 $\pm$ 0.16	85.50 $\pm$ 0.12	89.21 $\pm$ 0.06	90.29 $\pm$ 0.01	92.36 $\pm$ 0.18	84.43 $\pm$ 1.43
	w/	67.33 $\pm$ 0.82 (+1.02)	96.44 $\pm$ 0.22 (+1.11)	89.22 $\pm$ 0.62 (+3.72)	90.43 $\pm$ 0.44 (+1.22)	91.39 $\pm$ 0.23 (+1.10)	93.54 $\pm$ 0.07 (+1.18)	85.56 $\pm$ 0.36 (+1.13)
RAC-LoRA	w/o	58.11 $\pm$ 0.83	94.30 $\pm$ 0.05	75.73 $\pm$ 0.12	89.92 $\pm$ 0.06	89.52 $\pm$ 0.02	92.46 $\pm$ 0.13	73.54 $\pm$ 1.72
	w/	59.19 $\pm$ 1.38 (+1.08)	95.50 $\pm$ 0.19 (+1.20)	80.02 $\pm$ 0.56 (+4.29)	90.83 $\pm$ 0.65 (+0.91)	90.63 $\pm$ 0.16 (+1.11)	94.51 $\pm$ 0.32 (+2.05)	75.08 $\pm$ 0.61 (+1.54)

Table 8. Test accuracy(%) of Llama-3.1-8B with LoRA variants (with/without ETF) on GLUE benchmark datasets (rank 16).

Method	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
LoRA	w/o	67.11 $\pm$ 0.52	96.90 $\pm$ 0.11	89.13 $\pm$ 0.23	90.80 $\pm$ 0.14	91.25 $\pm$ 0.05	95.47 $\pm$ 0.06	85.44 $\pm$ 0.85
	w/	69.73 $\pm$ 0.49 (+2.62)	96.62 $\pm$ 0.18 (-0.28)	89.46 $\pm$ 0.25 (+0.33)	92.07 $\pm$ 0.08 (+1.27)	91.28 $\pm$ 0.06 (+0.03)	95.77 $\pm$ 0.08 (+0.30)	89.17 $\pm$ 0.90 (+3.73)
LoRA-GA	w/o	67.10 $\pm$ 0.11	96.56 $\pm$ 0.23	89.22 $\pm$ 0.49	91.12 $\pm$ 0.10	91.10 $\pm$ 0.02	94.99 $\pm$ 0.01	88.09 $\pm$ 0.36
	w/	69.73 $\pm$ 0.49 (+2.63)	96.62 $\pm$ 0.18 (+0.06)	89.46 $\pm$ 0.25 (+0.24)	92.07 $\pm$ 0.08 (+0.95)	91.28 $\pm$ 0.06 (+0.18)	95.77 $\pm$ 0.08 (+0.78)	89.17 $\pm$ 0.90 (+1.08)
PiSSA	w/o	65.57 $\pm$ 0.97	96.90 $\pm$ 0.23	88.73 $\pm$ 0.29	90.94 $\pm$ 0.01	91.59 $\pm$ 0.07	95.41 $\pm$ 0.08	86.82 $\pm$ 1.26
	w/	69.73 $\pm$ 0.49 (+4.16)	96.62 $\pm$ 0.18 (-0.28)	89.46 $\pm$ 0.25 (+0.73)	92.07 $\pm$ 0.08 (+1.13)	91.28 $\pm$ 0.06 (-0.31)	95.77 $\pm$ 0.08 (+0.36)	89.17 $\pm$ 0.90 (+2.35)
FLORA	w/o	67.14 $\pm$ 0.82	95.99 $\pm$ 0.05	87.01 $\pm$ 0.49	90.25 $\pm$ 0.02	90.01 $\pm$ 0.03	93.16 $\pm$ 0.13	85.92 $\pm$ 0.75
	w/	68.16 $\pm$ 0.61 (+1.02)	96.10 $\pm$ 0.03 (+0.11)	87.75 $\pm$ 0.98 (+0.74)	91.29 $\pm$ 0.02 (+1.04)	91.23 $\pm$ 0.01 (+1.22)	95.64 $\pm$ 0.01 (+2.48)	86.21 $\pm$ 0.46 (+0.29)
AsymLoRA	w/o	62.60 $\pm$ 0.92	95.91 $\pm$ 0.05	78.92 $\pm$ 1.23	89.66 $\pm$ 0.03	90.91 $\pm$ 0.07	93.92 $\pm$ 0.04	75.29 $\pm$ 0.36
	w/	65.14 $\pm$ 0.87 (+2.54)	96.11 $\pm$ 0.06 (+0.20)	81.13 $\pm$ 1.10 (+2.21)	90.02 $\pm$ 0.04 (+0.36)	91.34 $\pm$ 0.18 (+0.43)	95.11 $\pm$ 0.44 (+1.19)	77.25 $\pm$ 0.72 (+1.96)
ReLoRA	w/o	68.57 $\pm$ 0.48	96.27 $\pm$ 0.17	89.09 $\pm$ 0.37	89.66 $\pm$ 0.09	90.93 $\pm$ 0.05	94.46 $\pm$ 0.03	87.18 $\pm$ 0.90
	w/	67.53 $\pm$ 0.47 (-1.04)	96.67 $\pm$ 0.11 (+0.40)	89.21 $\pm$ 0.37 (+0.12)	90.47 $\pm$ 0.18 (+0.81)	91.07 $\pm$ 0.02 (+0.14)	95.05 $\pm$ 0.01 (+0.59)	87.72 $\pm$ 0.36 (+0.54)
RAC-LoRA	w/o	63.85 $\pm$ 0.59	95.93 $\pm$ 0.06	78.93 $\pm$ 1.47	89.11 $\pm$ 0.06	90.58 $\pm$ 0.02	93.87 $\pm$ 0.07	73.97 $\pm$ 0.18
	w/	64.92 $\pm$ 0.61 (+1.07)	96.34 $\pm$ 0.43 (+0.41)	79.90 $\pm$ 0.25 (+0.97)	89.56 $\pm$ 0.03 (+0.45)	90.65 $\pm$ 0.03 (+0.07)	95.01 $\pm$ 0.39 (+1.14)	76.89 $\pm$ 1.44 (+2.92)

Table 9. Test accuracy(%) of Llama-3.1-8B with Galore (with/without ETF) on GLUE benchmark datasets.

Rank	ETF	CoLA	SST-2	MRPC	QQP	MNLI	QNLI	RTE
4	w/o	63.12 $\pm$ 1.73	96.84 $\pm$ 0.29	84.18 $\pm$ 0.91	90.57 $\pm$ 0.97	90.06 $\pm$ 0.05	94.78 $\pm$ 0.07	87.91 $\pm$ 0.91
	w/	65.18 $\pm$ 2.15 (+2.06)	96.36 $\pm$ 0.17 (-0.48)	87.99 $\pm$ 0.74 (+3.81)	91.72 $\pm$ 0.11 (+1.15)	90.60 $\pm$ 0.06 (+0.54)	95.02 $\pm$ 0.06 (+0.24)	89.35 $\pm$ 1.61 (+1.44)
8	w/o	66.90 $\pm$ 1.93	96.13 $\pm$ 0.45	88.46 $\pm$ 1.33	90.25 $\pm$ 0.71	90.16 $\pm$ 0.10	94.40 $\pm$ 0.28	87.99 $\pm$ 1.46
	w/	69.32 $\pm$ 1.66 (+2.42)	97.25 $\pm$ 0.72 (+1.12)	90.20 $\pm$ 0.85 (+1.74)	91.34 $\pm$ 0.08 (+1.09)	91.43 $\pm$ 0.12 (+1.27)	95.60 $\pm$ 0.03 (+1.20)	89.14 $\pm$ 1.30 (+1.15)
16	w/o	62.43 $\pm$ 1.65	95.70 $\pm$ 0.06	87.14 $\pm$ 0.71	90.89 $\pm$ 0.51	88.58 $\pm$ 0.33	94.77 $\pm$ 0.09	88.39 $\pm$ 0.77
	w/	62.69 $\pm$ 1.79 (+0.26)	96.67 $\pm$ 0.36 (+0.97)	88.63 $\pm$ 0.77 (+1.49)	91.82 $\pm$ 0.12 (+0.93)	89.67 $\pm$ 0.03 (+1.09)	95.06 $\pm$ 0.03 (+0.29)	88.55 $\pm$ 0.93 (+0.16)

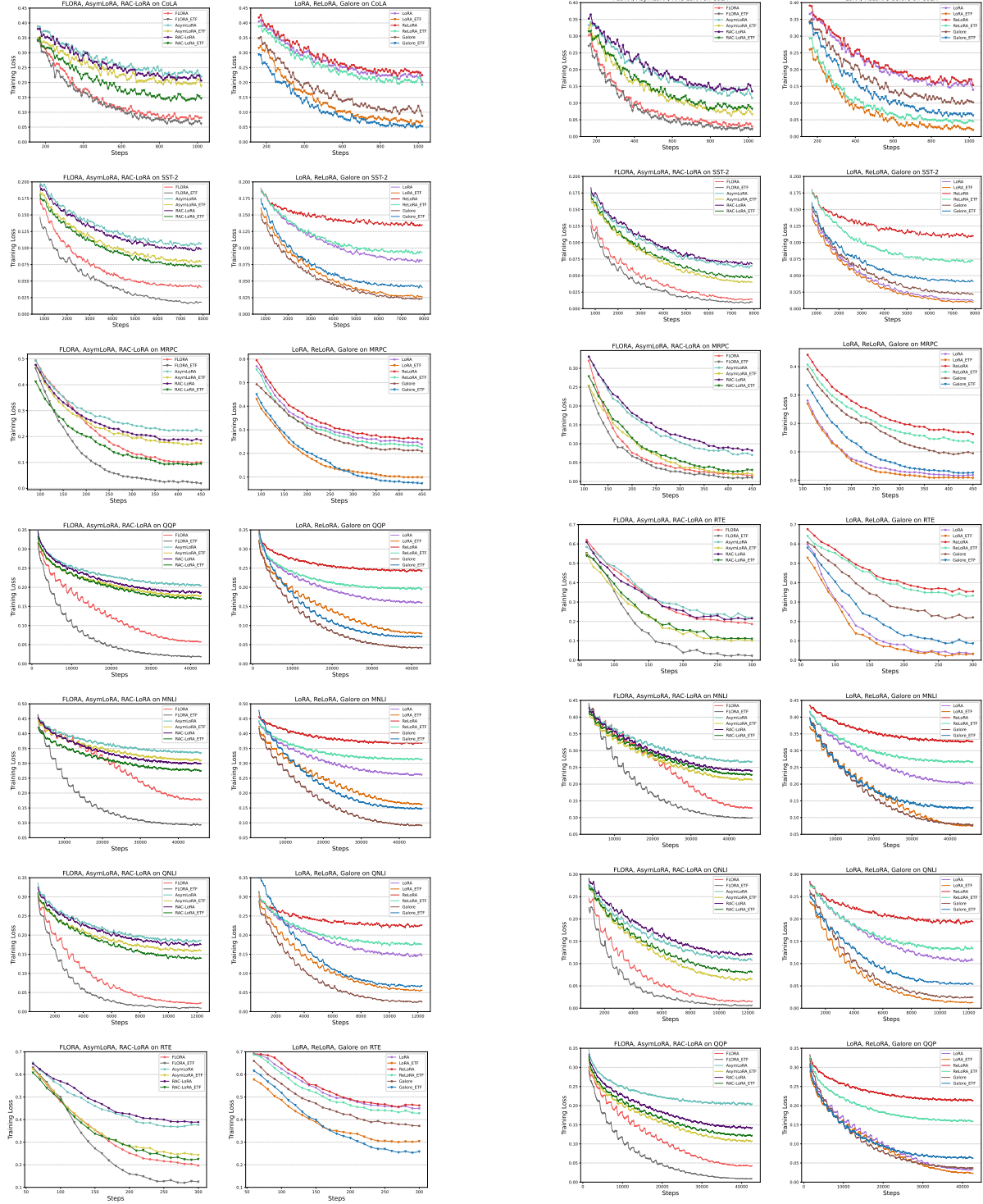


Figure 3. Convergence performance of LoRA variants with and without ETF on GLUE benchmark datasets, with the rank 4.

Figure 4. Convergence performance of LoRA variants with and without ETF on GLUE benchmark datasets, with the rank 8.

B.3. Natural Language Generation Tasks

B.3.1. SETUP

For natural language generation tasks, we evaluate these methods on GSM8K (Cobbe et al., 2021), HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021), and MT-Bench (Zheng et al., 2024a) benchmarks, including math, code, and instruction-following tasks. For the GSM8K task, we train the model on a 100k subset of the MetaMathQA dataset (Yu et al., 2024) and report the accuracy on GSM8K. For the HumanEval and MBPP tasks, we train the model on a 100k subset of the Code-Feedback dataset (Zheng et al., 2024b) and report the PASS@1 metric on HumanEval and MBPP. For the MT-Bench task, we train the model on a 52k subset of the WizardLM-Evol-Instruct dataset (Xu et al., 2024) and report the response quality score on MT-Bench.

We use the Llama-3.1-8B model (Dubey et al., 2024) and the Mistral-7B-v0.1 (Jiang et al., 2023) model implemented by the HuggingFace Transformers library (Wolf et al., 2020) as the backbone. For both models, we use the FP16/BF16 mixed-precision strategy to reduce the GPU memory consumption. All the experiments are conducted on a platform with 8 NVIDIA GeForce RTX 4090 GPUs.

For the Llama-3.1-8B model and Mistral-7B-v0.1 model, we adopt the AdamW optimizer (Loshchilov & Hutter, 2019), and report the best result with the learning rate $\eta \in \{1 \times 10^{-3}, 5 \times 10^{-4}, 1 \times 10^{-4}, 5 \times 10^{-5}, 1 \times 10^{-5}\}$ with cosine annealing for the learning rate schedule. We apply the LoRA adapters to multi-head attention layers following (Hu et al., 2022). The batch size is 1, the number of epochs is 3, the gradient accumulation step is 8, the scaling factor is twice the rank, and the maximum sequence length is 1024.

B.3.2. ADDITIONAL RESULTS

We repeat all the experiments over 5 times and report the average score with standard deviation of the Llama-3.1-8B and Mistral-7B-v0.1 model for rank 4, 8, and 16 in Table 10, Table 3, and Table 11. The results show that these LoRA variants with ETF consistently outperform their default initialization counterparts on natural language generation tasks.

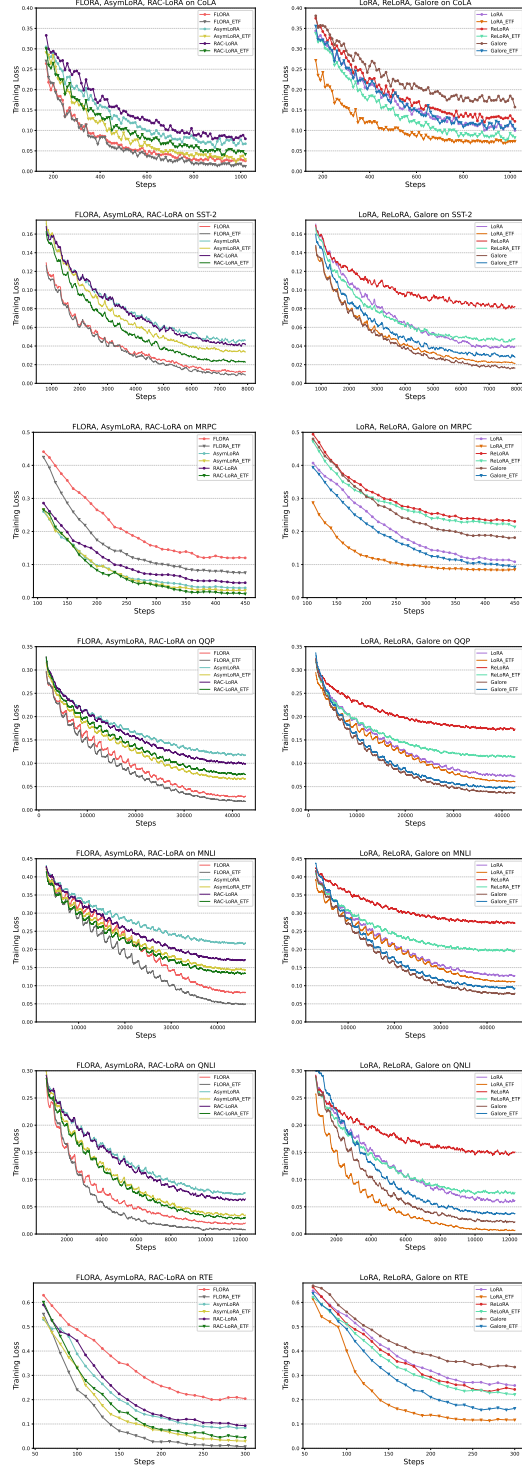


Figure 5. Convergence performance of LoRA variants with and without ETF on GLUE benchmark datasets, with the rank 16.

Towards Understanding the Dynamics of Low-Rank Adaptation

Table 10. Test score(%) of Llama-3.1-8B and Mistral-7B-v0.1 with LoRA variants (with/without ETF) on NLG datasets (rank 4).

Method	ETF	Llama-3.1-8B				Mistral-7B-v0.1			
		GSM8K	HumanEval	MBPP	MT-Bench	GSM8K	HumanEval	MBPP	MT-Bench
LoRA	w/o	64.40 $\pm$ 0.04	43.49 $\pm$ 1.15	50.68 $\pm$ 0.19	6.13 $\pm$ 0.05	64.94 $\pm$ 1.25	29.93 $\pm$ 1.22	40.17 $\pm$ 0.77	6.04 $\pm$ 0.09
	w/	66.79 $\pm$ 0.61 (+2.39)	44.21 $\pm$ 0.91 (+0.72)	51.03 $\pm$ 0.05 (+0.35)	6.23 $\pm$ 0.04 (+0.10)	68.34 $\pm$ 0.30 (+3.40)	30.89 $\pm$ 0.52 (+0.96)	41.55 $\pm$ 0.10 (+1.38)	6.24 $\pm$ 0.01 (+0.20)
LoRA-GA	w/o	65.28 $\pm$ 0.04	43.72 $\pm$ 1.44	50.39 $\pm$ 0.72	6.21 $\pm$ 0.06	69.11 $\pm$ 0.34	30.59 $\pm$ 0.61	41.41 $\pm$ 0.36	6.12 $\pm$ 0.05
	w/	66.79 $\pm$ 0.61 (+1.51)	44.21 $\pm$ 0.91 (+0.49)	51.03 $\pm$ 0.05 (+0.64)	6.23 $\pm$ 0.04 (+0.02)	68.34 $\pm$ 0.30 (-0.77)	30.89 $\pm$ 0.52 (+0.30)	41.55 $\pm$ 0.10 (+0.14)	6.24 $\pm$ 0.01 (+0.12)
PiSSA	w/o	65.02 $\pm$ 0.04	43.53 $\pm$ 0.30	50.62 $\pm$ 0.49	6.25 $\pm$ 0.03	64.56 $\pm$ 0.34	30.80 $\pm$ 0.31	40.37 $\pm$ 0.97	6.09 $\pm$ 0.02
	w/	66.79 $\pm$ 0.61 (+1.77)	44.21 $\pm$ 0.91 (+0.68)	51.03 $\pm$ 0.05 (+0.41)	6.23 $\pm$ 0.04 (-0.02)	68.34 $\pm$ 0.30 (+3.78)	30.89 $\pm$ 0.52 (+0.09)	41.55 $\pm$ 0.10 (+1.18)	6.24 $\pm$ 0.01 (+0.15)
FLORA	w/o	67.24 $\pm$ 0.56	43.41 $\pm$ 0.98	46.67 $\pm$ 0.36	5.92 $\pm$ 0.15	68.43 $\pm$ 0.41	36.80 $\pm$ 1.31	41.51 $\pm$ 0.16	5.98 $\pm$ 0.03
	w/	70.74 $\pm$ 0.38 (+3.50)	44.82 $\pm$ 0.91 (+1.41)	47.45 $\pm$ 0.41 (+0.78)	5.96 $\pm$ 0.09 (+0.04)	67.47 $\pm$ 0.38 (-0.96)	37.25 $\pm$ 1.60 (+0.45)	40.71 $\pm$ 0.05 (-0.80)	6.01 $\pm$ 0.04 (+0.03)
AsymLoRA	w/o	58.36 $\pm$ 0.34	39.48 $\pm$ 1.00	50.80 $\pm$ 0.20	5.96 $\pm$ 0.07	58.11 $\pm$ 0.41	28.80 $\pm$ 0.31	40.78 $\pm$ 0.26	6.01 $\pm$ 0.05
	w/	60.57 $\pm$ 0.65 (+2.21)	43.19 $\pm$ 0.71 (+3.71)	51.47 $\pm$ 0.06 (+0.67)	6.01 $\pm$ 0.01 (+0.05)	60.69 $\pm$ 0.04 (+2.58)	28.93 $\pm$ 0.61 (+0.13)	41.04 $\pm$ 0.41 (+0.26)	6.13 $\pm$ 0.02 (+0.12)
ReLoRA	w/o	63.05 $\pm$ 0.25	43.59 $\pm$ 0.30	50.70 $\pm$ 0.41	5.94 $\pm$ 0.06	62.05 $\pm$ 0.85	30.49 $\pm$ 0.61	40.94 $\pm$ 0.62	6.05 $\pm$ 0.01
	w/	66.61 $\pm$ 0.27 (+3.56)	45.53 $\pm$ 1.04 (+1.94)	51.98 $\pm$ 0.20 (+1.28)	6.03 $\pm$ 0.02 (+0.09)	66.15 $\pm$ 0.19 (+4.10)	30.51 $\pm$ 1.91 (+0.02)	41.89 $\pm$ 0.26 (+0.95)	6.07 $\pm$ 0.03 (+0.02)
RAC-LoRA	w/o	57.21 $\pm$ 0.41	42.27 $\pm$ 0.58	50.57 $\pm$ 0.36	6.05 $\pm$ 0.02	57.17 $\pm$ 0.54	28.66 $\pm$ 1.22	41.08 $\pm$ 0.05	5.95 $\pm$ 0.03
	w/	59.46 $\pm$ 1.17 (+2.25)	43.22 $\pm$ 0.91 (+0.95)	51.11 $\pm$ 0.61 (+0.54)	5.97 $\pm$ 0.04 (-0.08)	61.83 $\pm$ 0.34 (+4.66)	29.67 $\pm$ 0.74 (+1.01)	40.48 $\pm$ 0.15 (-0.60)	6.06 $\pm$ 0.04 (+0.11)
GaLore	w/o	76.25 $\pm$ 0.72	48.17 $\pm$ 1.83	50.44 $\pm$ 0.46	6.25 $\pm$ 0.03	71.03 $\pm$ 0.19	42.11 $\pm$ 0.91	42.32 $\pm$ 1.23	6.23 $\pm$ 0.07
	w/	77.86 $\pm$ 0.73 (+1.61)	49.84 $\pm$ 1.89 (+1.67)	51.18 $\pm$ 0.31 (+0.74)	6.32 $\pm$ 0.06 (+0.07)	71.36 $\pm$ 0.04 (+0.33)	43.54 $\pm$ 1.22 (+1.43)	43.89 $\pm$ 0.26 (+1.57)	6.17 $\pm$ 0.05 (-0.06)

Table 11. Test score(%) of Llama-3.1-8B and Mistral-7B-v0.1 with LoRA variants (with/without ETF) on NLG datasets (rank 16).

Method	ETF	Llama-3.1-8B				Mistral-7B-v0.1			
		GSM8K	HumanEval	MBPP	MT-Bench	GSM8K	HumanEval	MBPP	MT-Bench
LoRA	w/o	68.83 $\pm$ 0.57	44.51 $\pm$ 1.49	51.87 $\pm$ 0.31	6.12 $\pm$ 0.05	70.39 $\pm$ 0.56	32.45 $\pm$ 0.30	42.28 $\pm$ 0.24	6.07 $\pm$ 0.06
	w/	71.01 $\pm$ 0.59 (+2.18)	46.95 $\pm$ 0.50 (+2.44)	53.16 $\pm$ 0.46 (+1.29)	6.37 $\pm$ 0.09 (+0.25)	70.93 $\pm$ 0.12 (+0.54)	34.28 $\pm$ 1.52 (+1.83)	42.95 $\pm$ 0.10 (+0.67)	6.34 $\pm$ 0.06 (+0.27)
LoRA-GA	w/o	70.82 $\pm$ 0.04	46.22 $\pm$ 1.83	52.72 $\pm$ 0.05	6.23 $\pm$ 0.04	70.85 $\pm$ 0.04	32.93 $\pm$ 0.61	42.77 $\pm$ 0.56	6.15 $\pm$ 0.07
	w/	71.01 $\pm$ 0.59 (+0.19)	46.95 $\pm$ 0.50 (+0.73)	53.16 $\pm$ 0.46 (+0.44)	6.37 $\pm$ 0.09 (+0.14)	70.93 $\pm$ 0.12 (+0.08)	34.28 $\pm$ 1.52 (+1.35)	42.95 $\pm$ 0.10 (+0.18)	6.34 $\pm$ 0.06 (+0.19)
PiSSA	w/o	69.06 $\pm$ 0.64	43.12 $\pm$ 0.91	52.64 $\pm$ 0.05	6.29 $\pm$ 0.07	68.88 $\pm$ 0.11	34.55 $\pm$ 0.21	42.89 $\pm$ 0.57	6.19 $\pm$ 0.04
	w/	71.01 $\pm$ 0.59 (+1.95)	46.95 $\pm$ 0.50 (+3.83)	53.16 $\pm$ 0.46 (+0.52)	6.37 $\pm$ 0.09 (+0.08)	70.93 $\pm$ 0.12 (+2.05)	34.28 $\pm$ 1.52 (-0.27)	42.95 $\pm$ 0.10 (+0.06)	6.34 $\pm$ 0.06 (+0.15)
FLORA	w/o	71.89 $\pm$ 0.68	46.04 $\pm$ 1.75	51.49 $\pm$ 0.46	6.08 $\pm$ 0.06	70.86 $\pm$ 0.91	39.95 $\pm$ 1.75	42.48 $\pm$ 0.88	6.08 $\pm$ 0.04
	w/	73.77 $\pm$ 0.23 (+1.88)	49.09 $\pm$ 0.30 (+3.05)	52.36 $\pm$ 0.41 (+0.87)	6.22 $\pm$ 0.04 (+0.14)	72.53 $\pm$ 0.58 (+1.67)	39.15 $\pm$ 1.13 (-0.80)	43.24 $\pm$ 1.24 (+0.76)	6.15 $\pm$ 0.05 (+0.07)
AsymLoRA	w/o	63.40 $\pm$ 0.51	42.88 $\pm$ 1.44	51.54 $\pm$ 0.36	6.16 $\pm$ 0.02	64.59 $\pm$ 0.92	30.71 $\pm$ 0.61	42.09 $\pm$ 0.25	6.04 $\pm$ 0.04
	w/	66.74 $\pm$ 0.52 (+3.34)	43.90 $\pm$ 0.50 (+1.02)	52.16 $\pm$ 0.20 (+0.62)	6.27 $\pm$ 0.03 (+0.11)	67.25 $\pm$ 0.38 (+2.66)	32.02 $\pm$ 0.30 (+1.31)	42.20 $\pm$ 0.46 (+0.11)	6.06 $\pm$ 0.21 (+0.02)
ReLoRA	w/o	66.43 $\pm$ 1.09	42.68 $\pm$ 0.50	51.69 $\pm$ 0.51	6.31 $\pm$ 0.04	68.34 $\pm$ 0.50	31.54 $\pm$ 1.22	42.35 $\pm$ 0.41	6.08 $\pm$ 0.13
	w/	69.54 $\pm$ 0.31 (+3.11)	46.34 $\pm$ 1.32 (+3.66)	52.92 $\pm$ 0.15 (+1.23)	6.27 $\pm$ 0.02 (-0.04)	69.94 $\pm$ 0.80 (+1.60)	33.37 $\pm$ 0.61 (+1.83)	42.89 $\pm$ 0.05 (+0.54)	6.28 $\pm$ 0.04 (+0.20)
RAC-LoRA	w/o	62.34 $\pm$ 0.84	42.45 $\pm$ 0.79	51.36 $\pm$ 0.36	6.08 $\pm$ 0.01	62.02 $\pm$ 0.30	30.62 $\pm$ 0.30	41.63 $\pm$ 0.30	6.02 $\pm$ 0.02
	w/	66.56 $\pm$ 0.66 (+4.22)	43.70 $\pm$ 0.29 (+1.25)	52.34 $\pm$ 0.05 (+0.98)	6.13 $\pm$ 0.05 (+0.05)	65.84 $\pm$ 1.09 (+3.82)	32.23 $\pm$ 0.91 (+1.61)	42.36 $\pm$ 0.21 (+0.73)	6.03 $\pm$ 0.10 (+0.01)
GaLore	w/o	76.91 $\pm$ 0.42	49.65 $\pm$ 1.04	52.73 $\pm$ 0.24	6.28 $\pm$ 0.07	71.27 $\pm$ 0.16	43.41 $\pm$ 0.92	46.92 $\pm$ 0.46	6.29 $\pm$ 0.04
	w/	78.12 $\pm$ 0.80 (+1.21)	51.42 $\pm$ 1.56 (+1.77)	53.39 $\pm$ 0.10 (+0.66)	6.42 $\pm$ 0.05 (+0.14)	73.77 $\pm$ 0.49 (+2.50)	46.03 $\pm$ 0.92 (+2.62)	47.05 $\pm$ 0.67 (+0.13)	6.17 $\pm$ 0.04 (-0.12)